

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

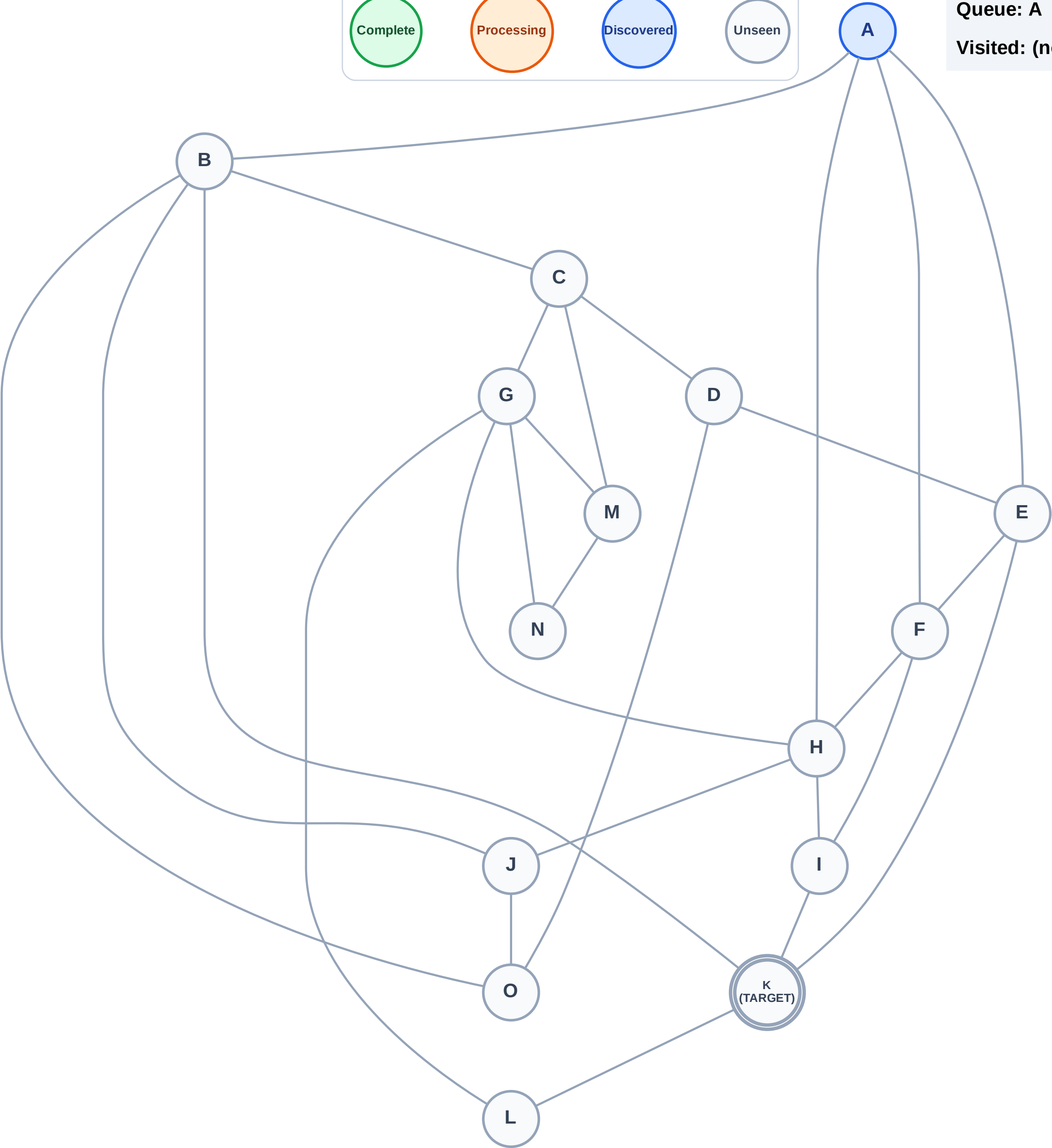
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

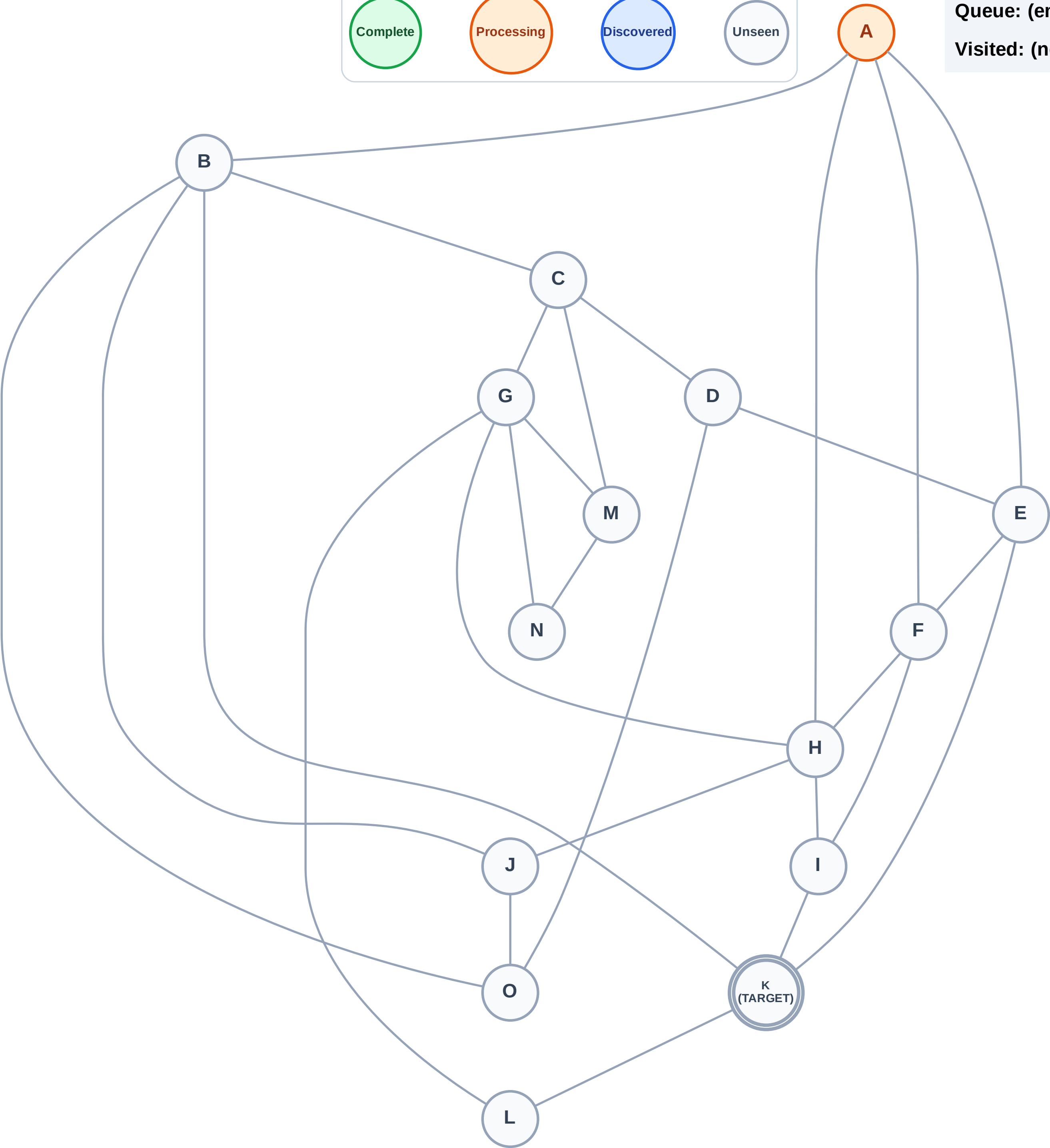
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



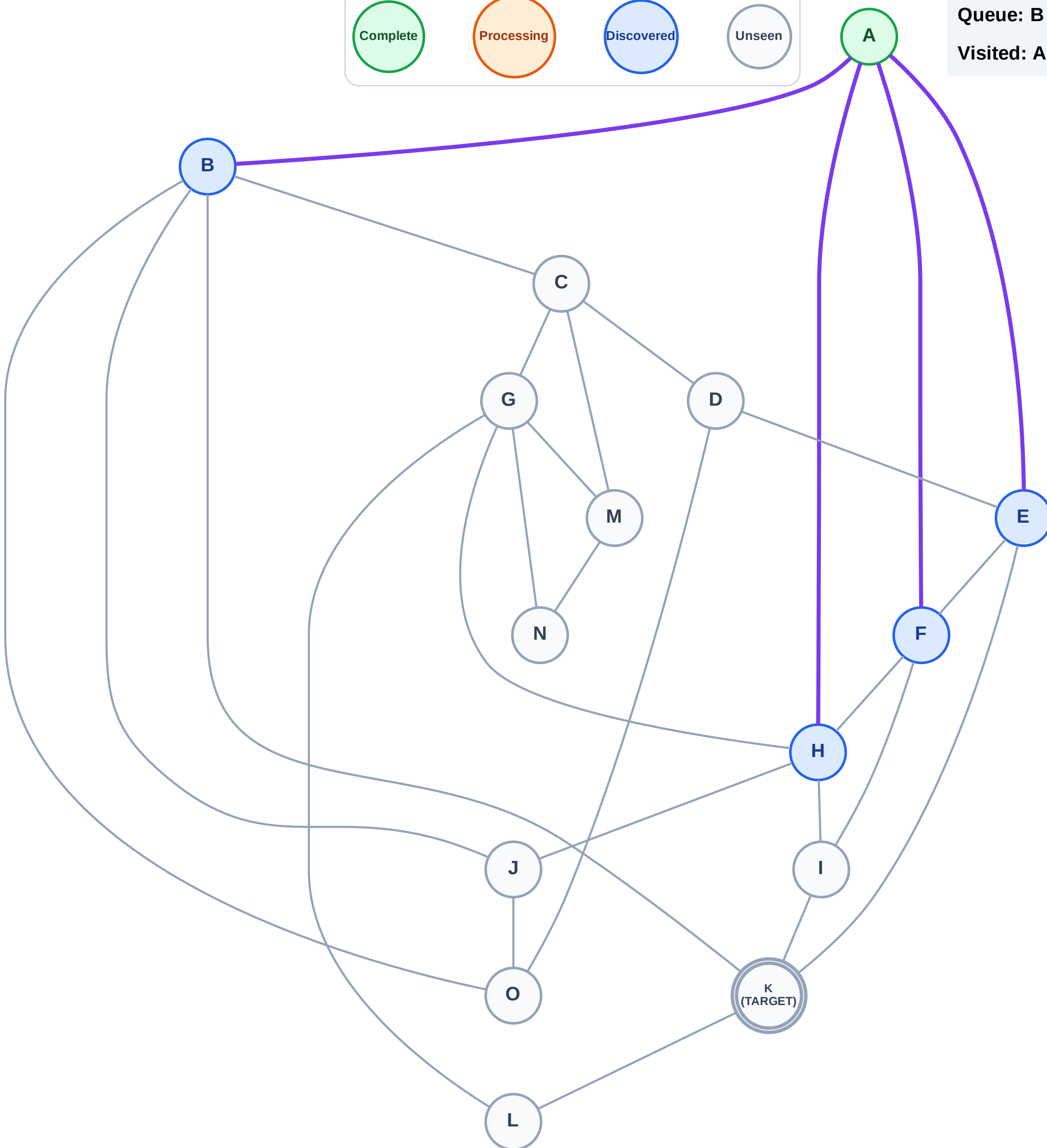
BFS Traversal

Step 3: Discover B, E, F, H from A

Legend



Queue: B → E → F → H
Visited: A



BFS Traversal

Step 4: Process node B

Legend

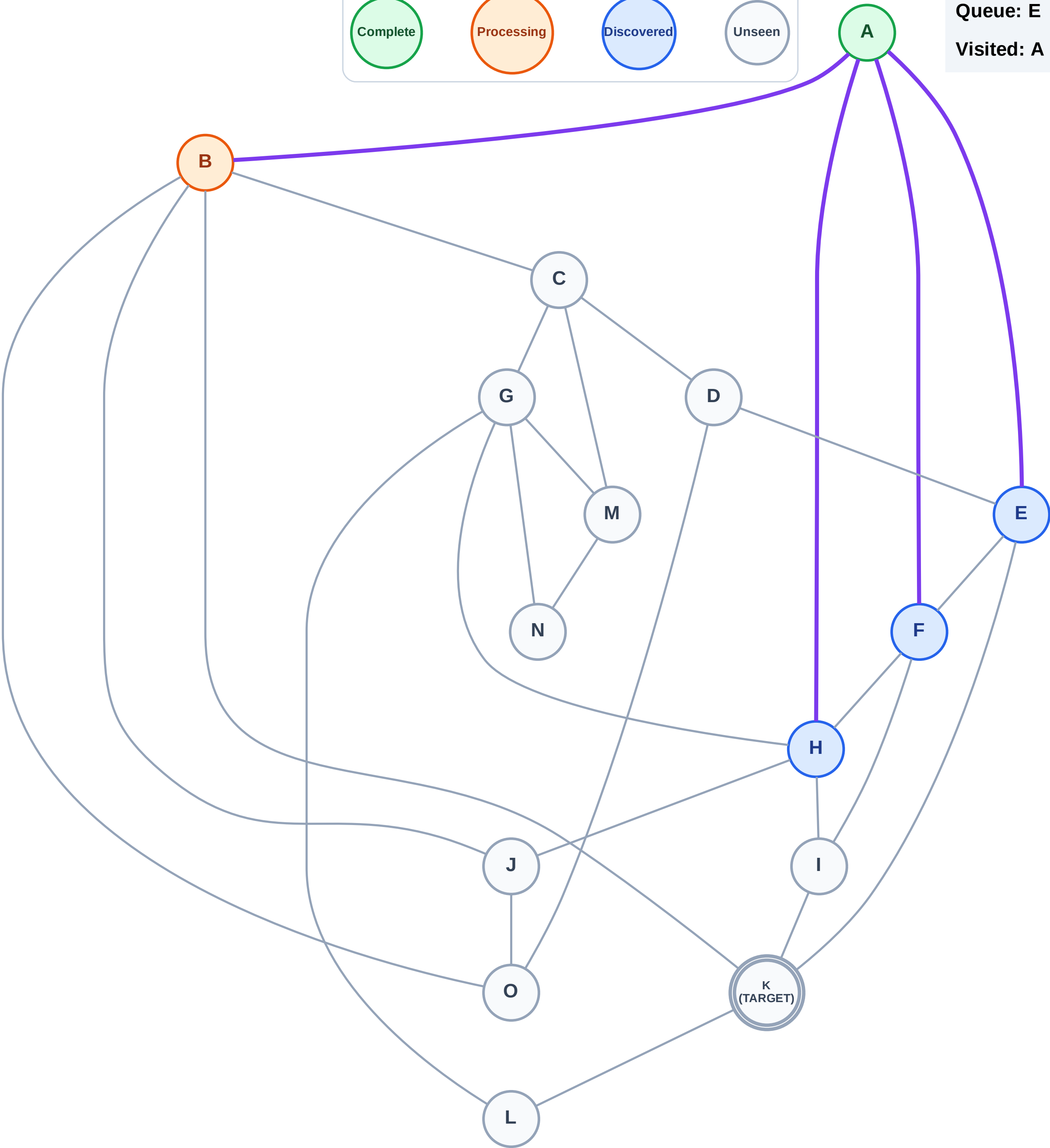
Complete

Processing

Discovered

Unseen

Queue: E → F → H
Visited: A



BFS Traversal

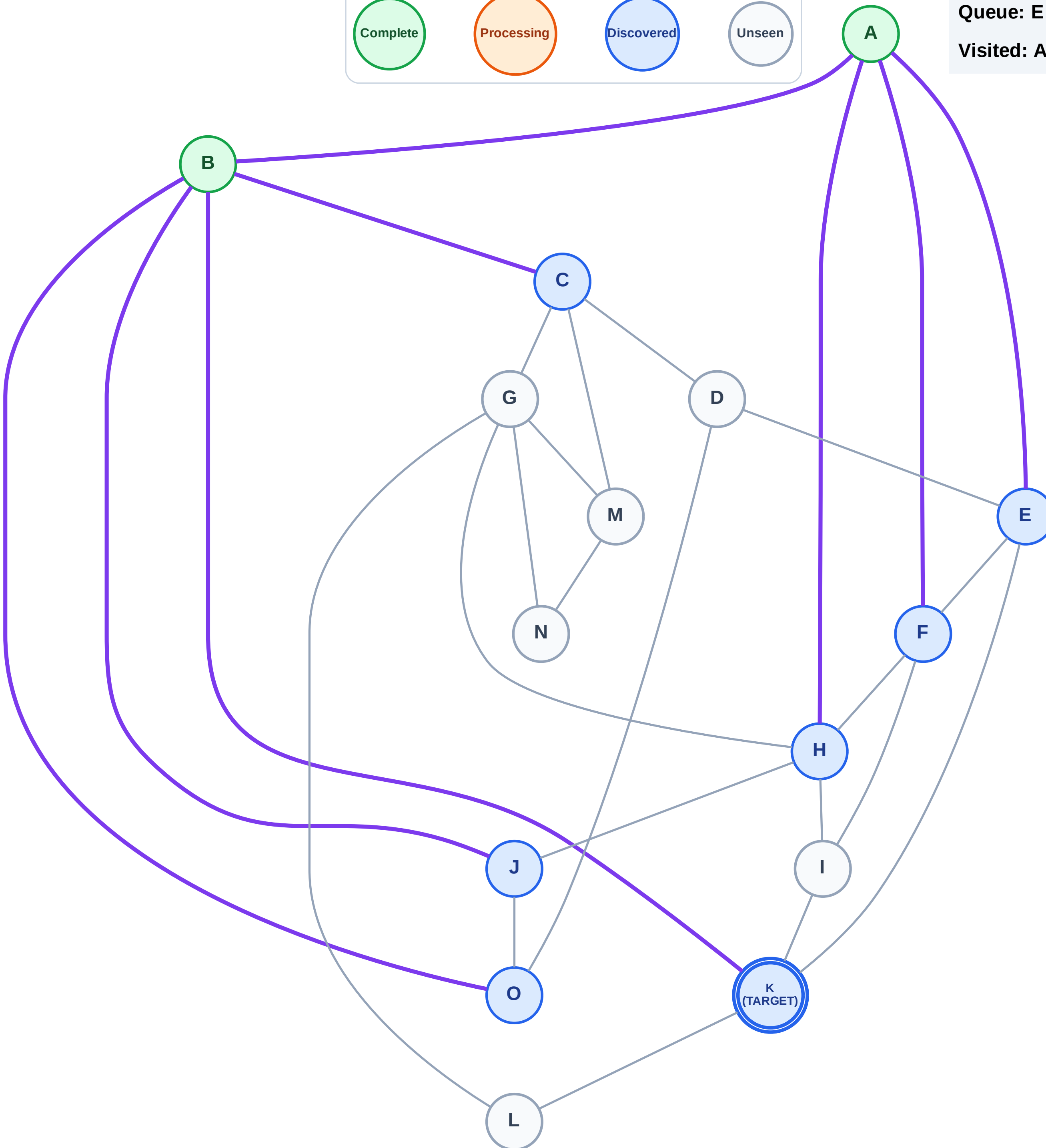
Step 5: Discover C, J, K, O from B

Legend



Queue: E → F → H → C → J → K → O

Visited: A, B



BFS Traversal

Step 6: Process node E

Legend



Complete



Processing



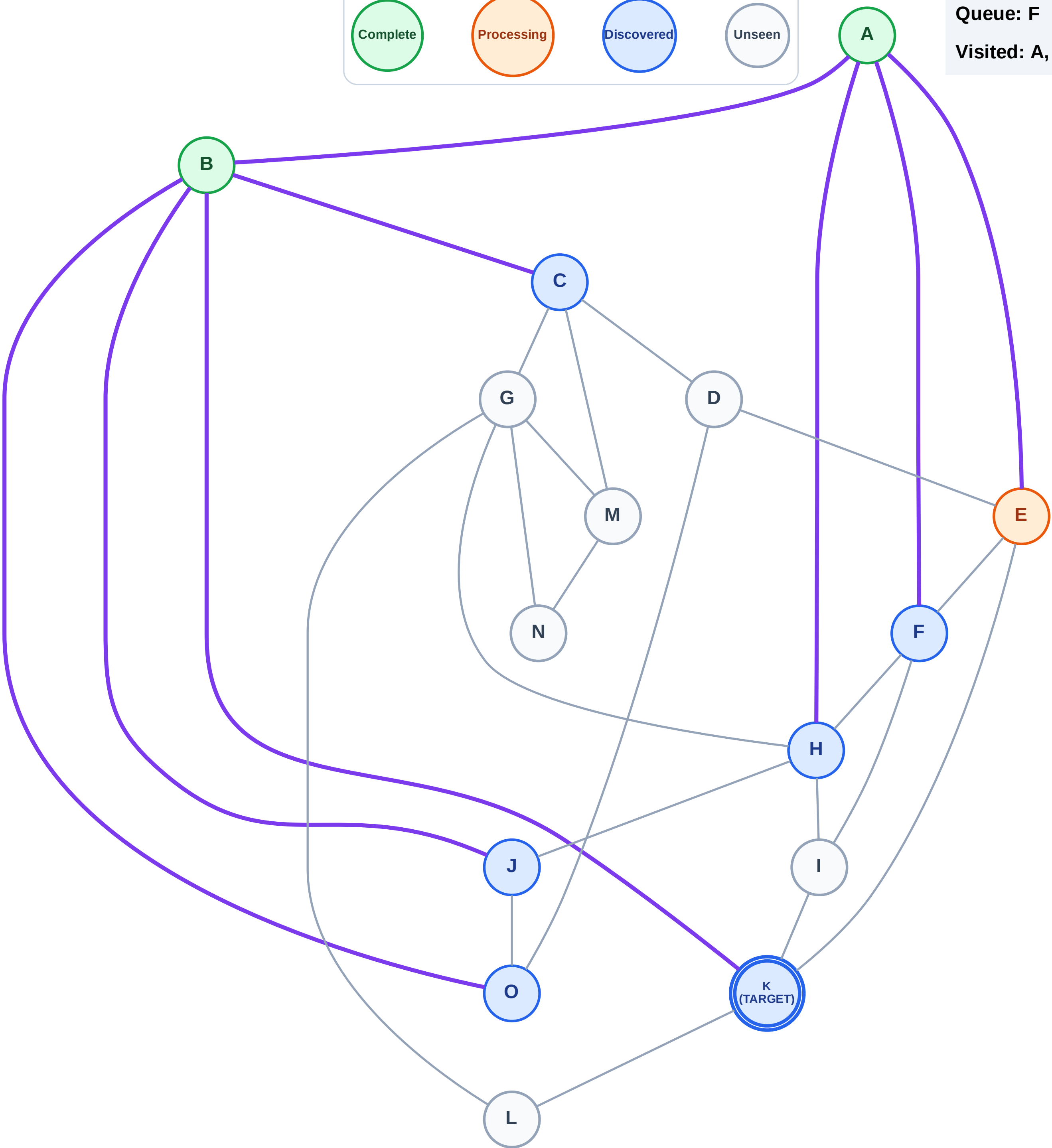
Discovered



Unseen

Queue: F → H → C → J → K → O

Visited: A, B



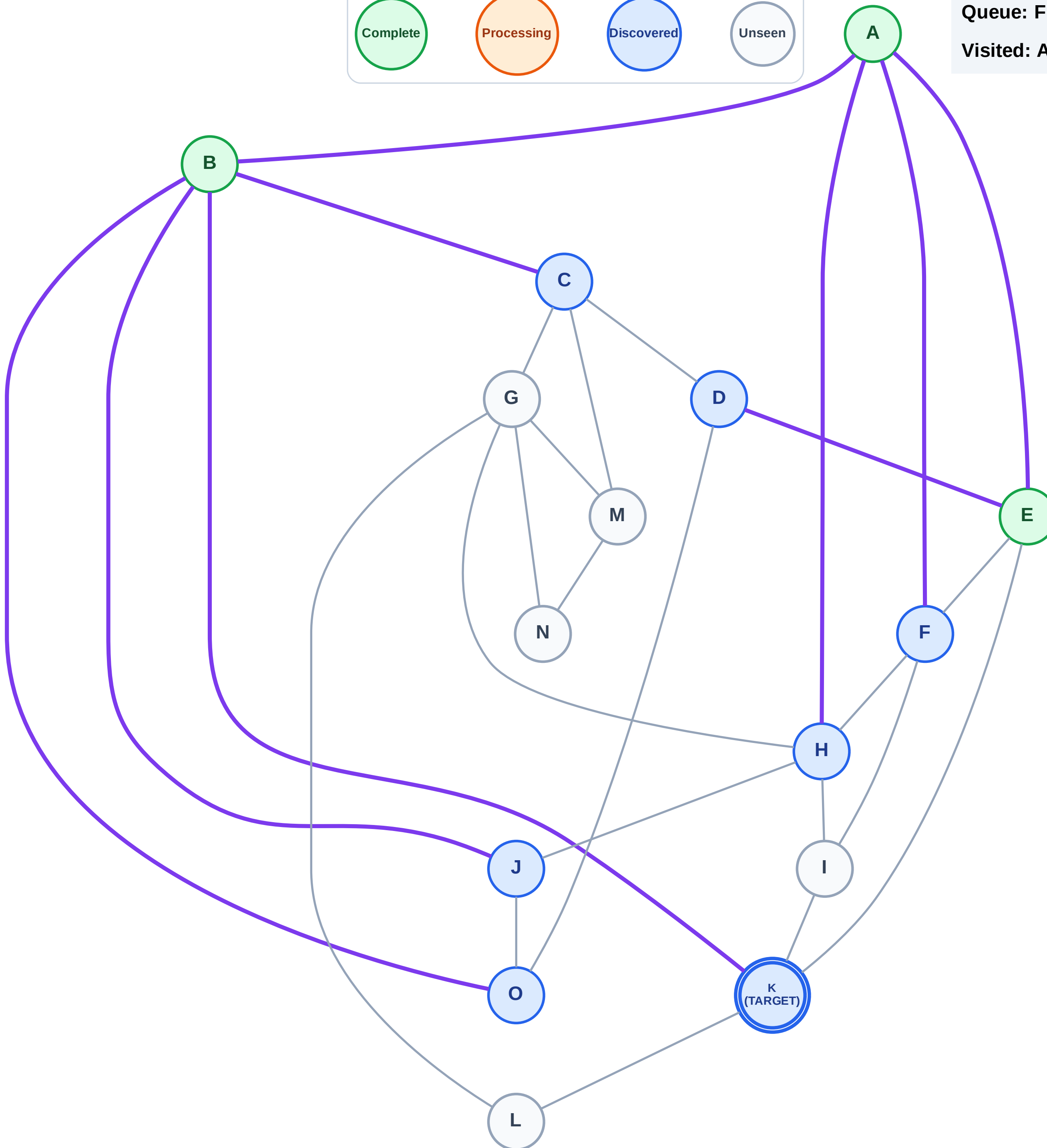
BFS Traversal

Step 7: Discover D from E

Legend



Queue: F → H → C → J → K → O → D
Visited: A, B, E



BFS Traversal

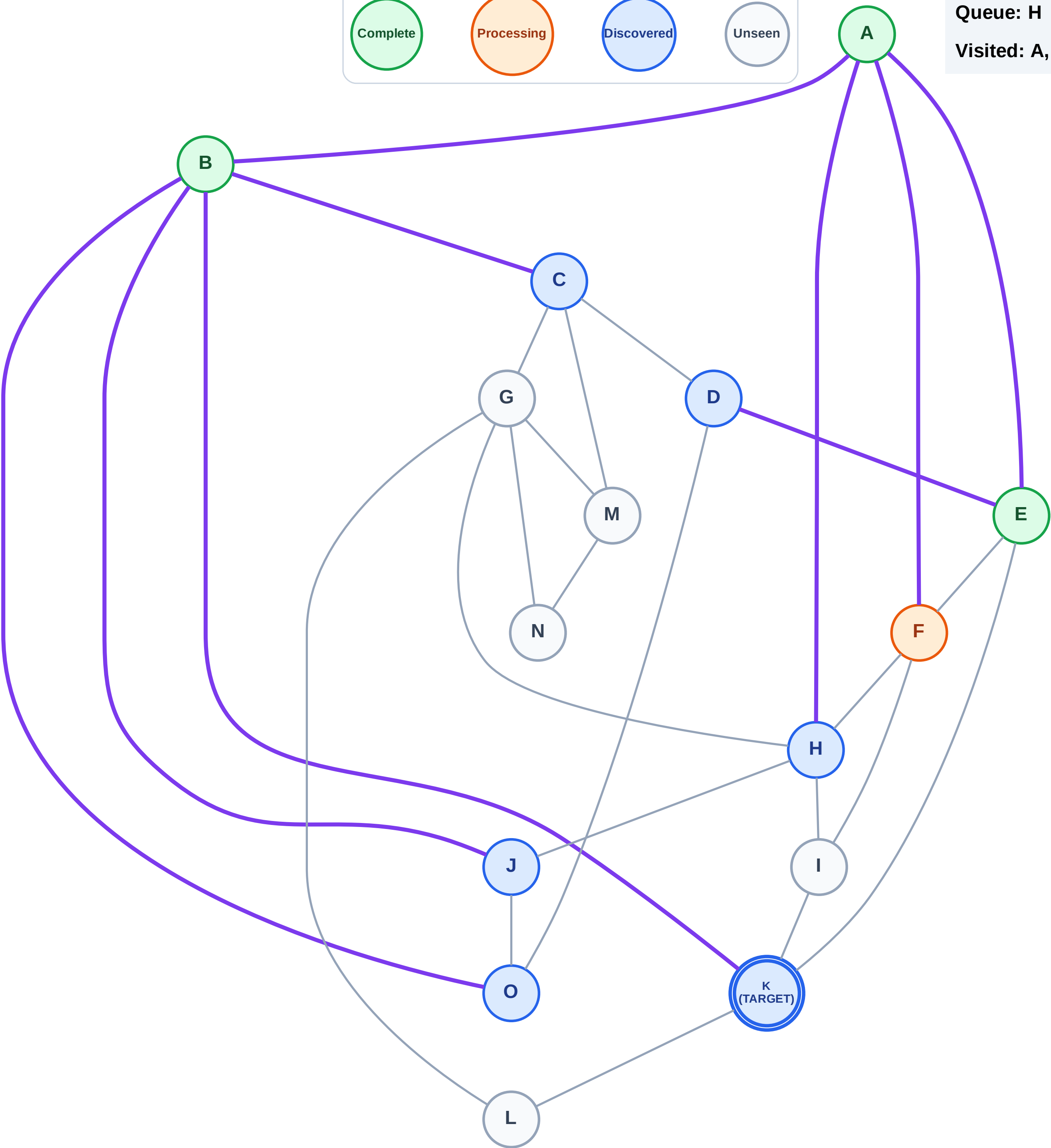
Step 8: Process node F

Legend



Queue: H → C → J → K → O → D

Visited: A, B, E



Step 9: Discover I from F

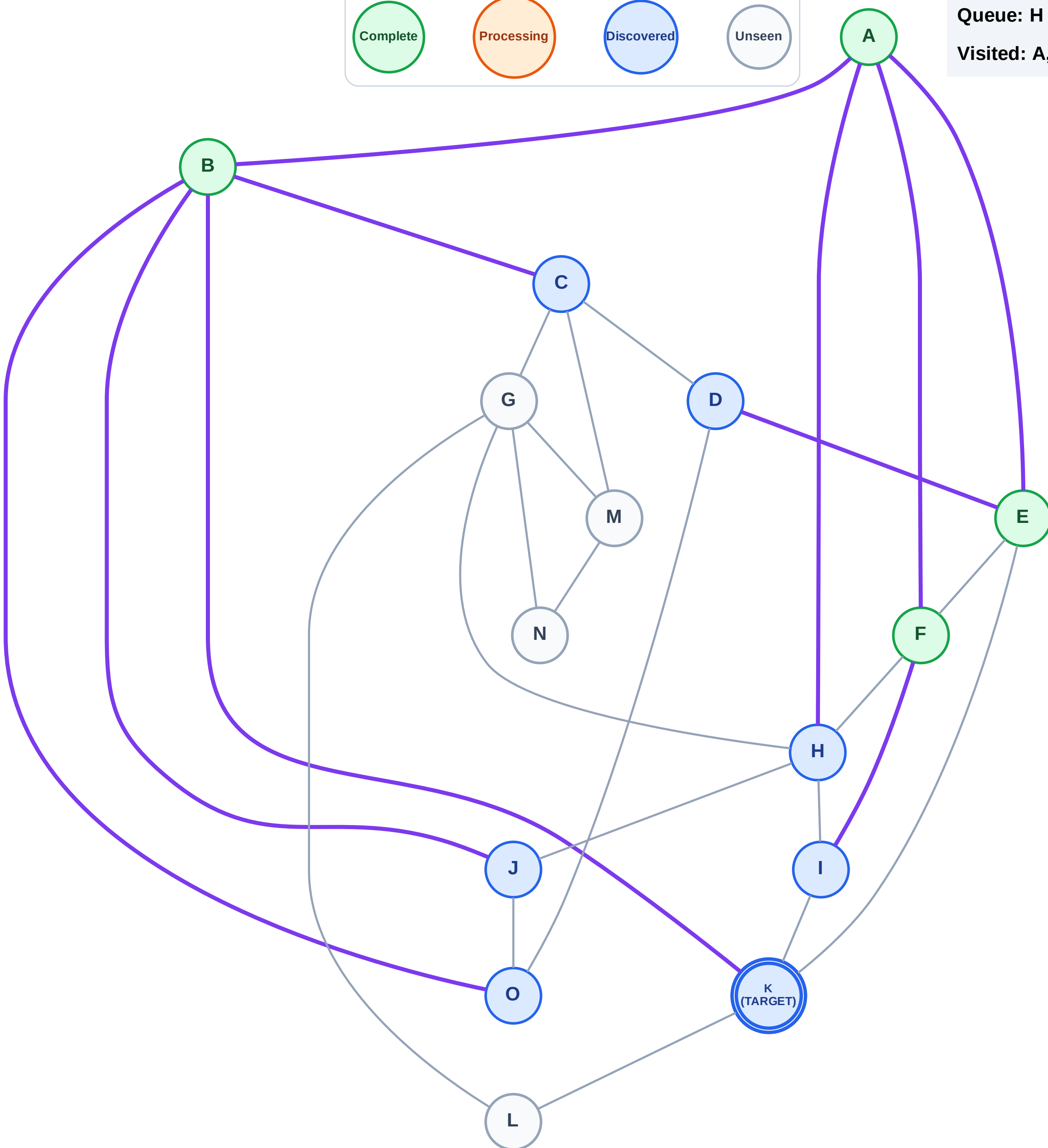
d

The diagram consists of four circles arranged horizontally. The first circle is green and labeled 'Complete'. The second circle is orange and labeled 'Processing'. The third circle is blue and labeled 'Discovered'. The fourth circle is light blue and labeled 'Unseen'.

Processing

Unseen

Visited: A, B, E, F



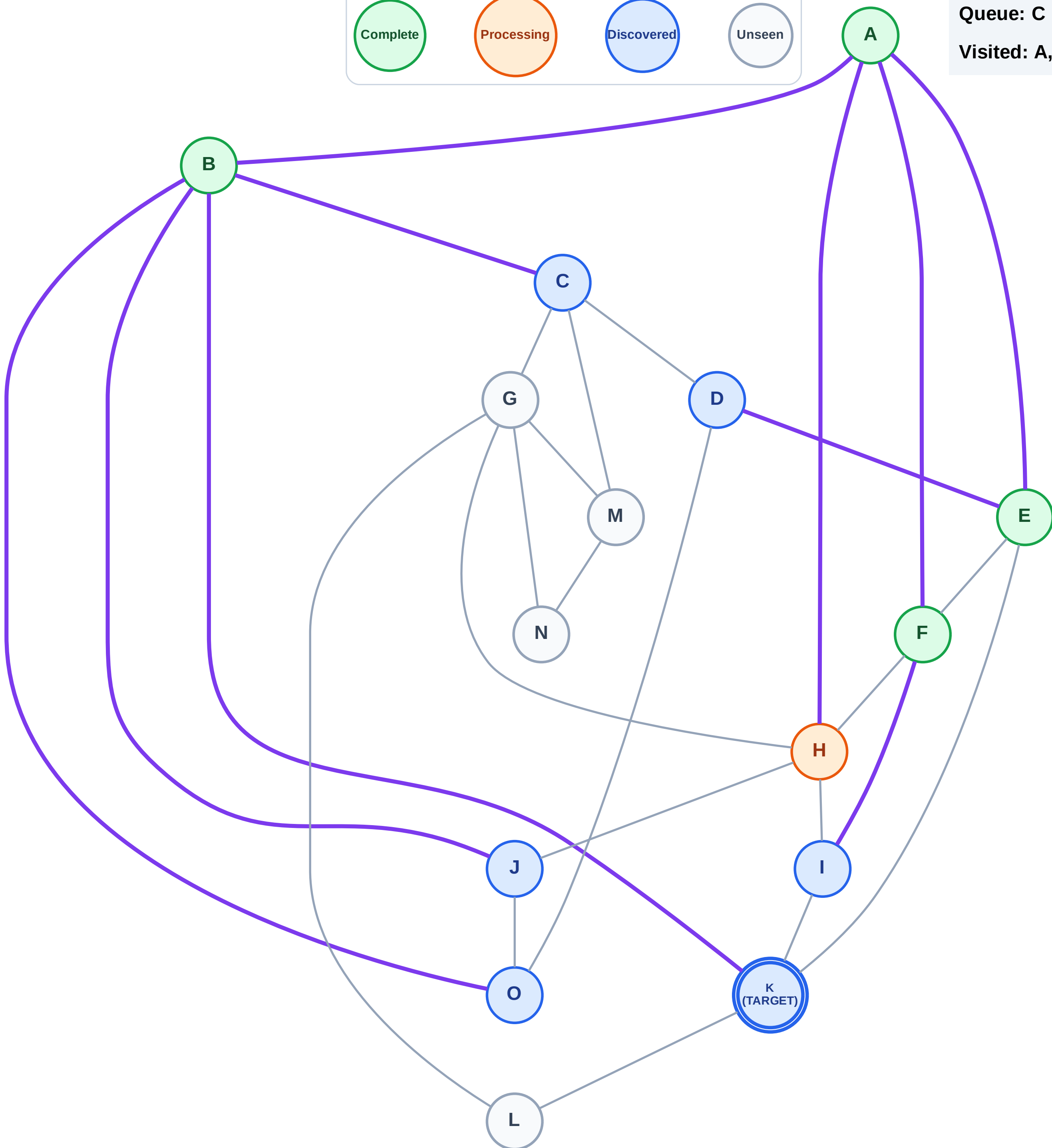
Step 10: Process node H

Legend

Complete



Visited: A, B, E, F



BFS Traversal

Step 11: Discover G from H

Legend



Complete



Processing

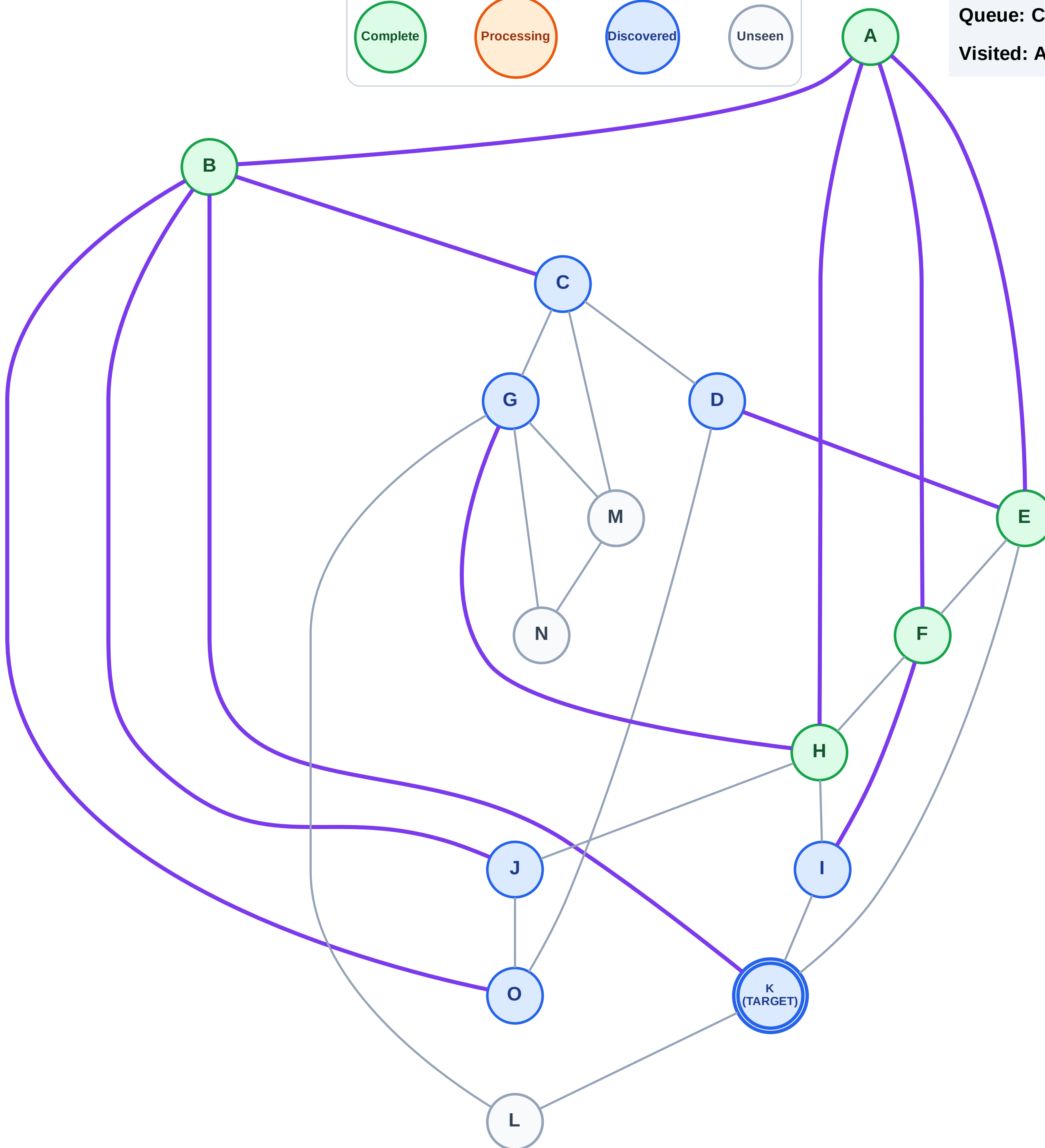


Discovered



Unseen

Queue: C → J → K → O → D → I → G
Visited: A, B, E, F, H



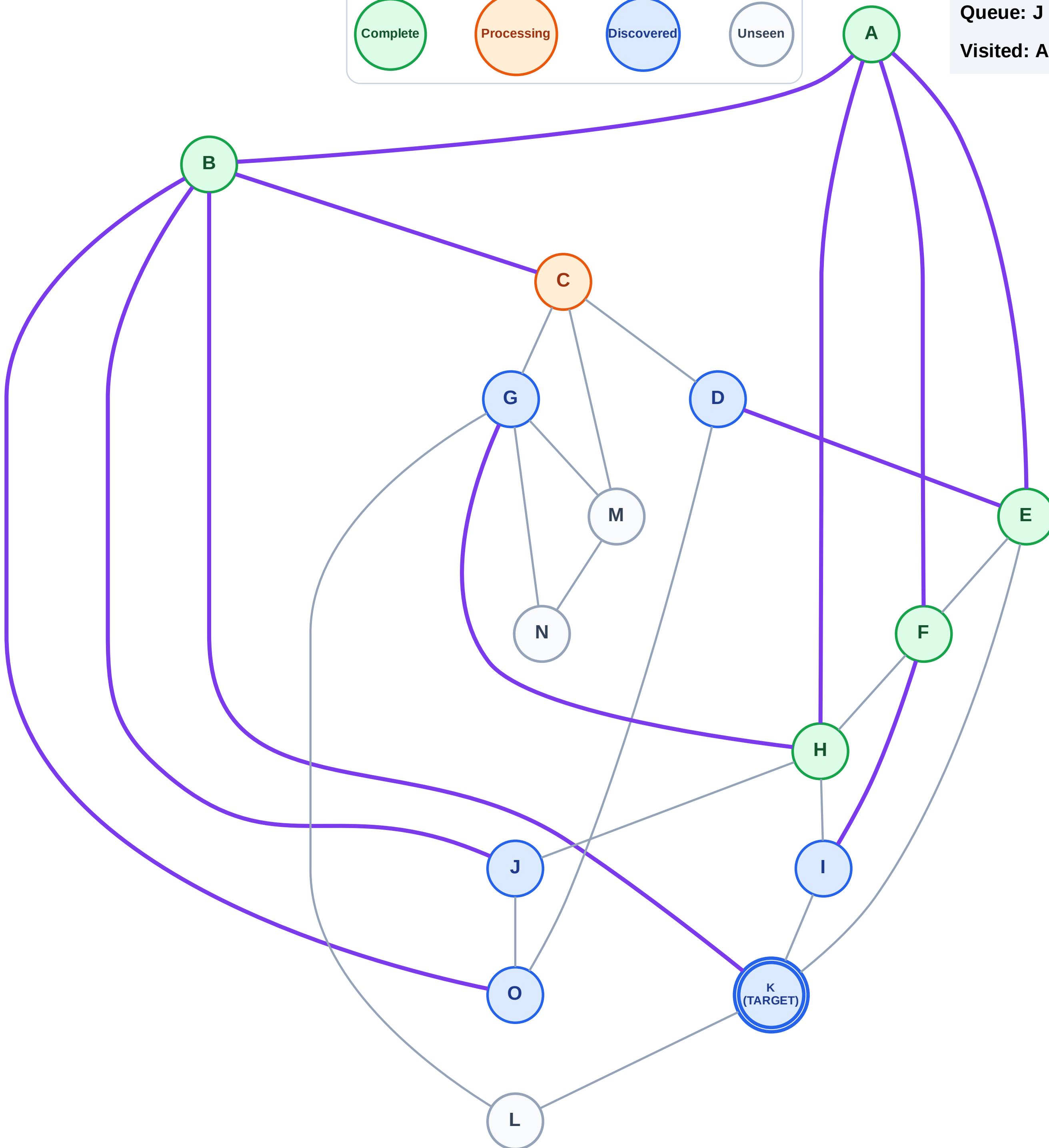
BFS Traversal

Step 12: Process node C

Legend



Queue: J → K → O → D → I → G
Visited: A, B, E, F, H



BFS Traversal

Step 13: Discover M from C

Legend



Complete



Processing

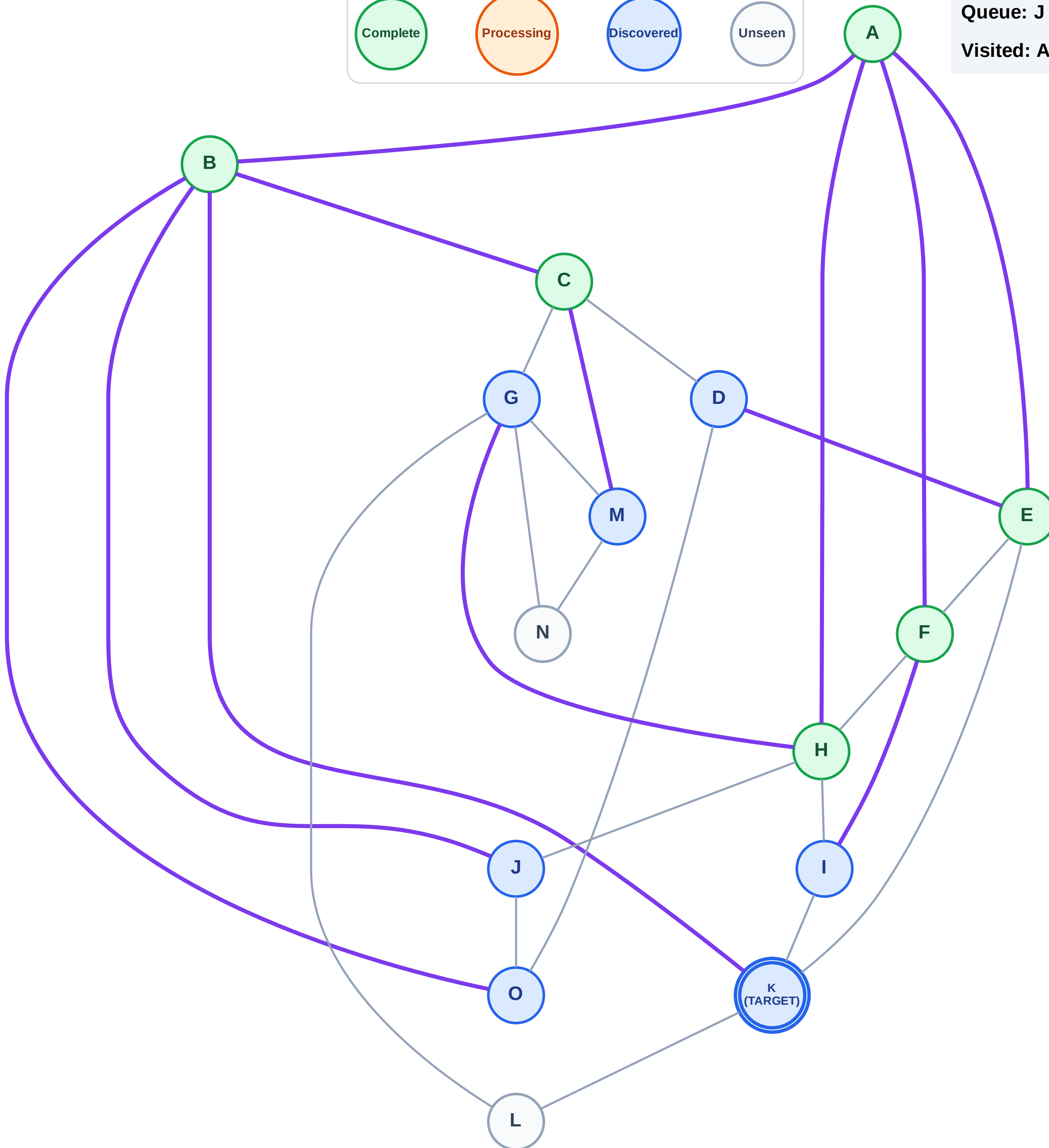


Discovered



Unseen

Queue: J → K → O → D → I → G → M
Visited: A, B, C, E, F, H



BFS Traversal

Step 14: Process node J

Legend



Complete



Processing



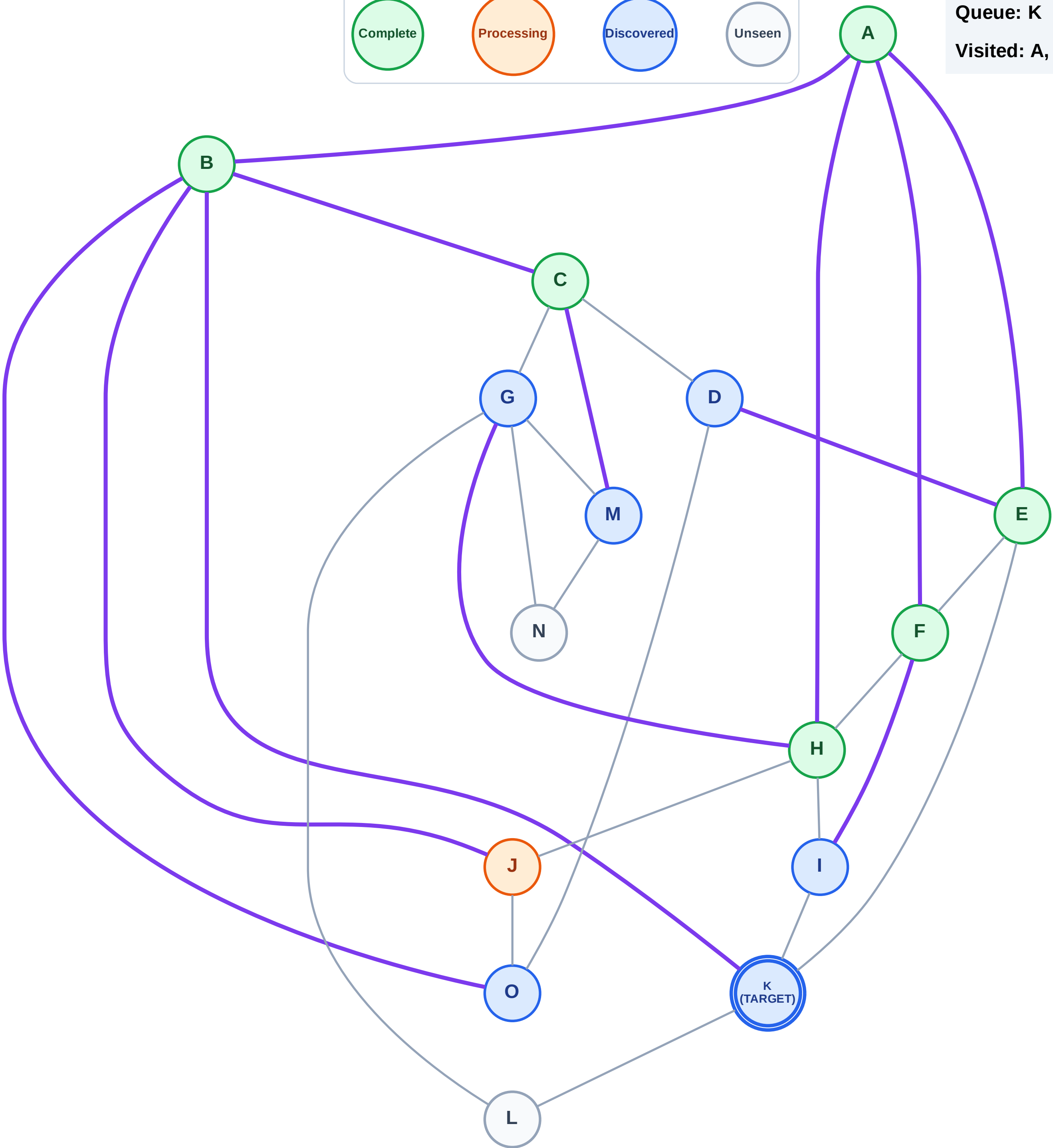
Discovered



Unseen

Queue: K → O → D → I → G → M

Visited: A, B, C, E, F, H



BFS Traversal

Step 15: No new neighbors from J

Legend



Complete



Processing

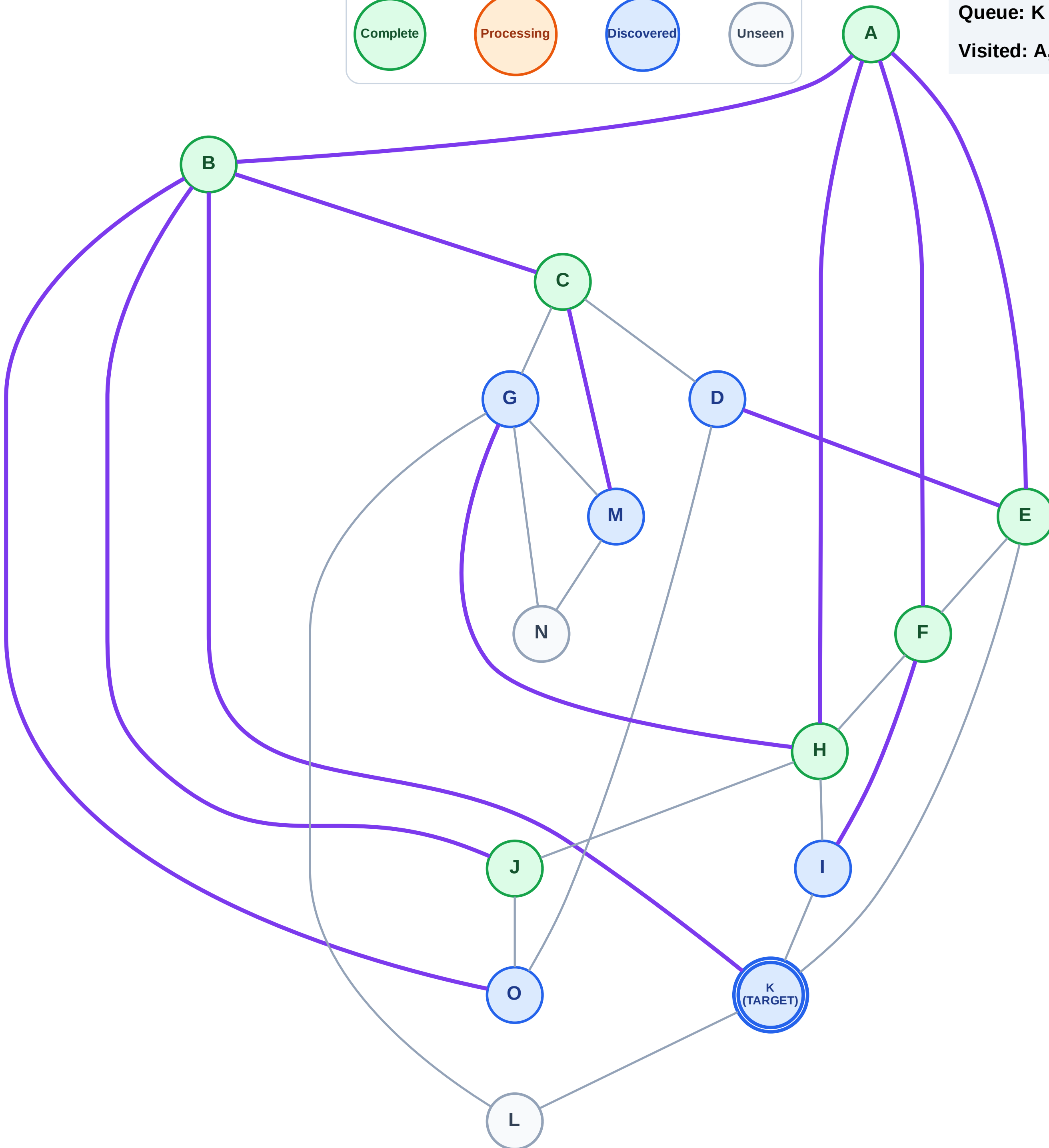


Discovered



Unseen

Queue: K → O → D → I → G → M
Visited: A, B, C, E, F, H, J



BFS Traversal

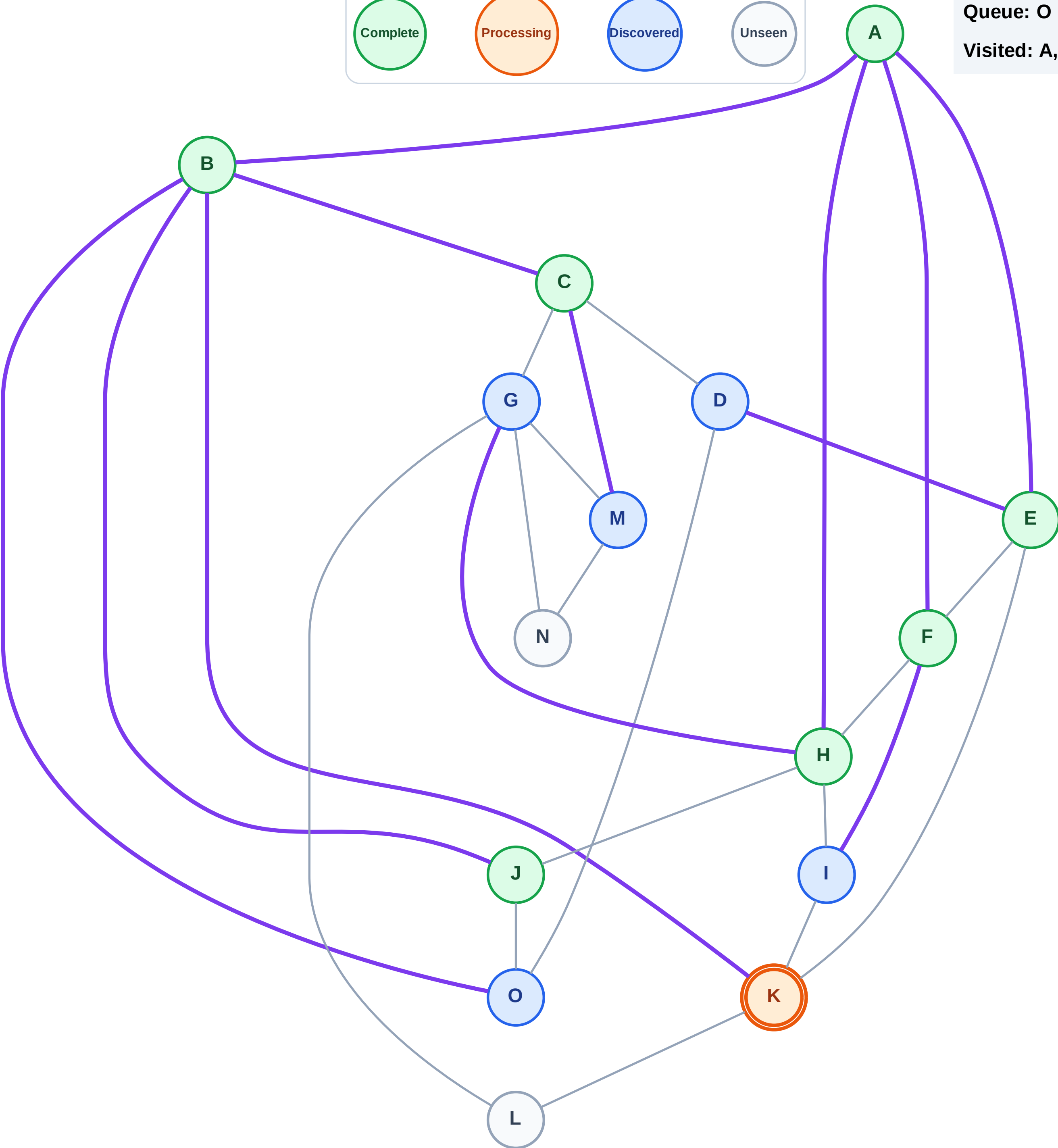
Step 16: Process node K

Legend



Queue: O → D → I → G → M

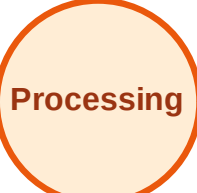
Visited: A, B, C, E, F, H, J



BFS Traversal

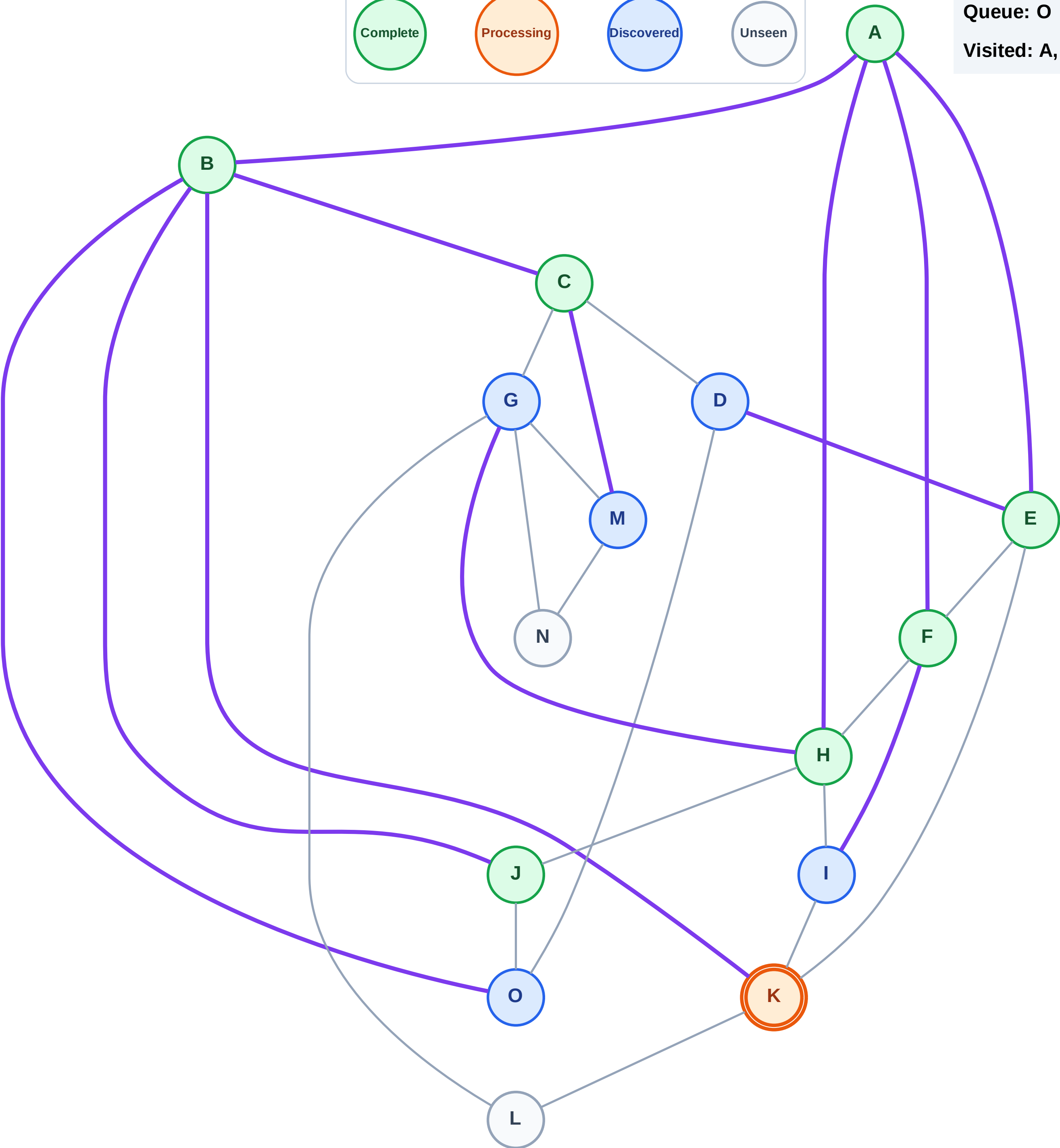
Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Queue: O → D → I → G → M

Visited: A, B, C, E, F, H, J, K



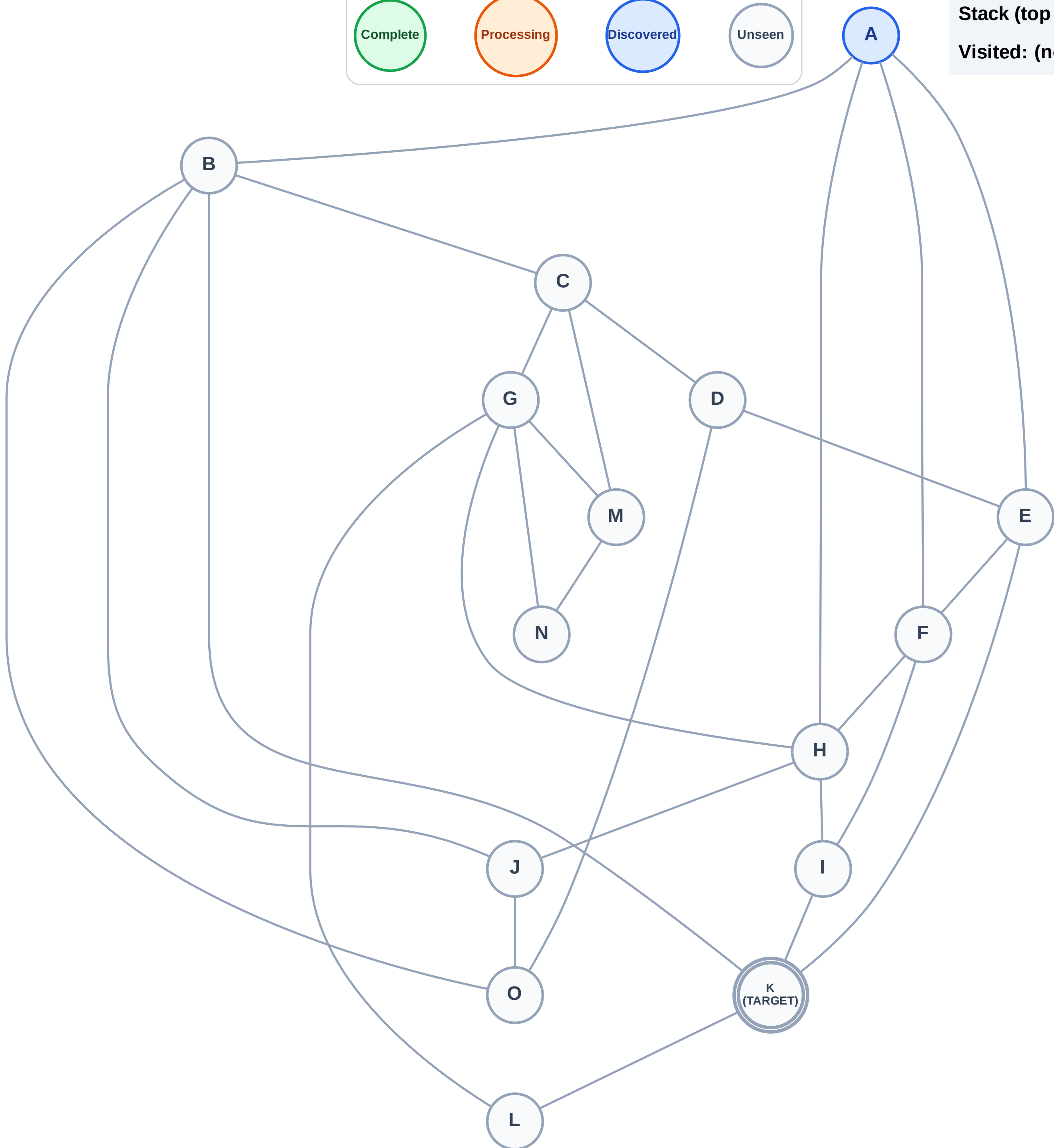
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

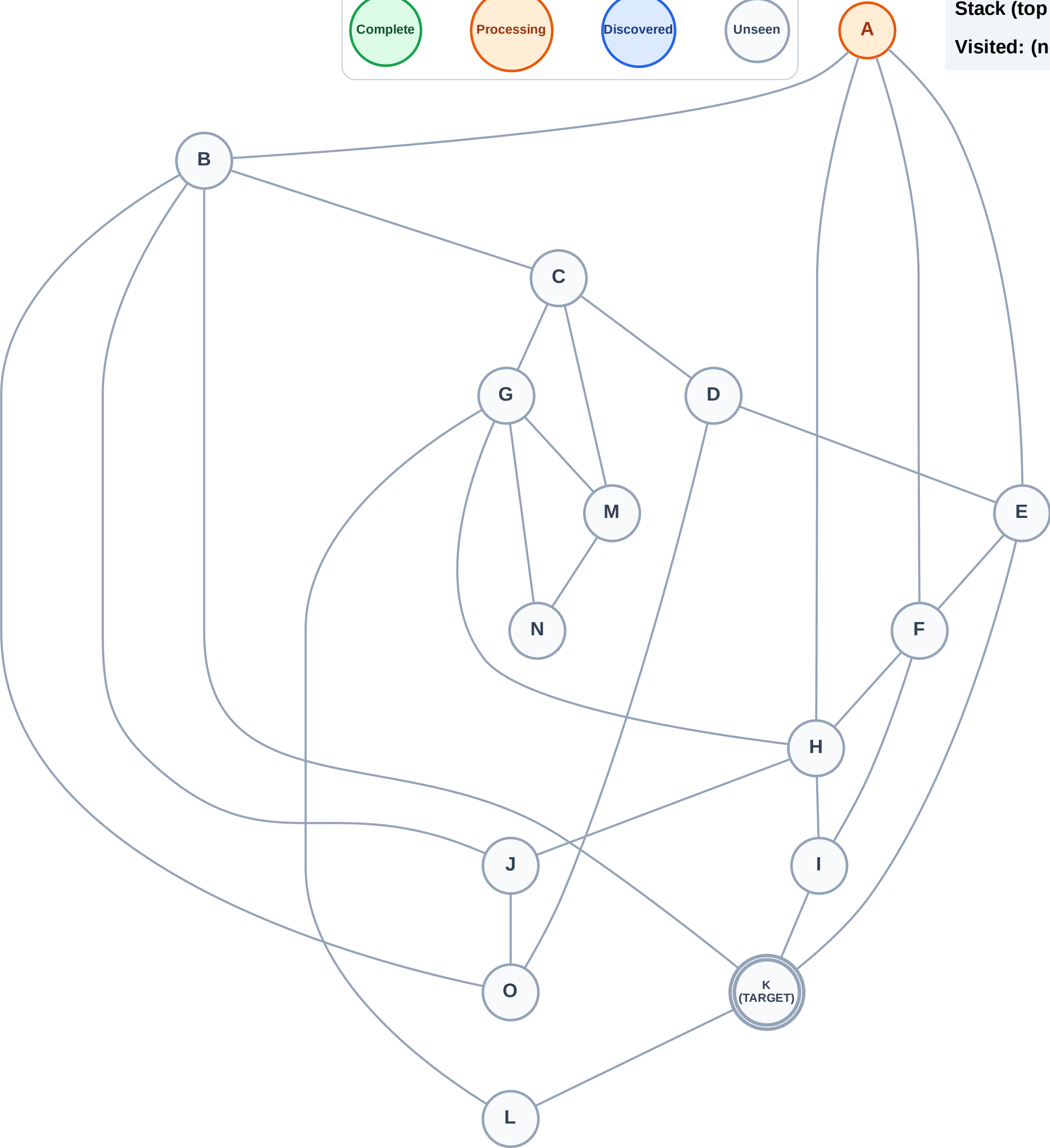
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)

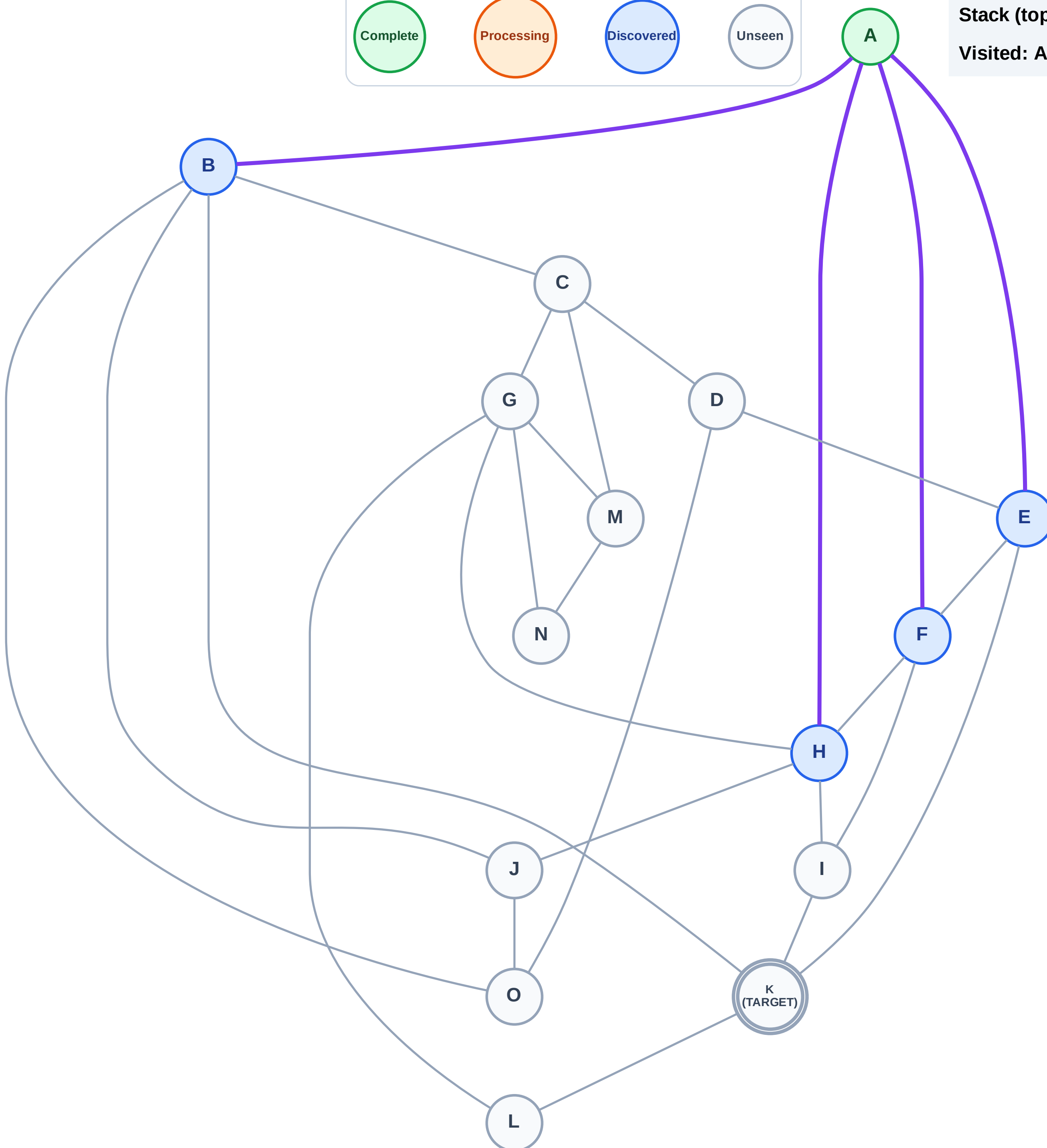


Step 3: Discover H, F, E, B from A

Legend

Unseen

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

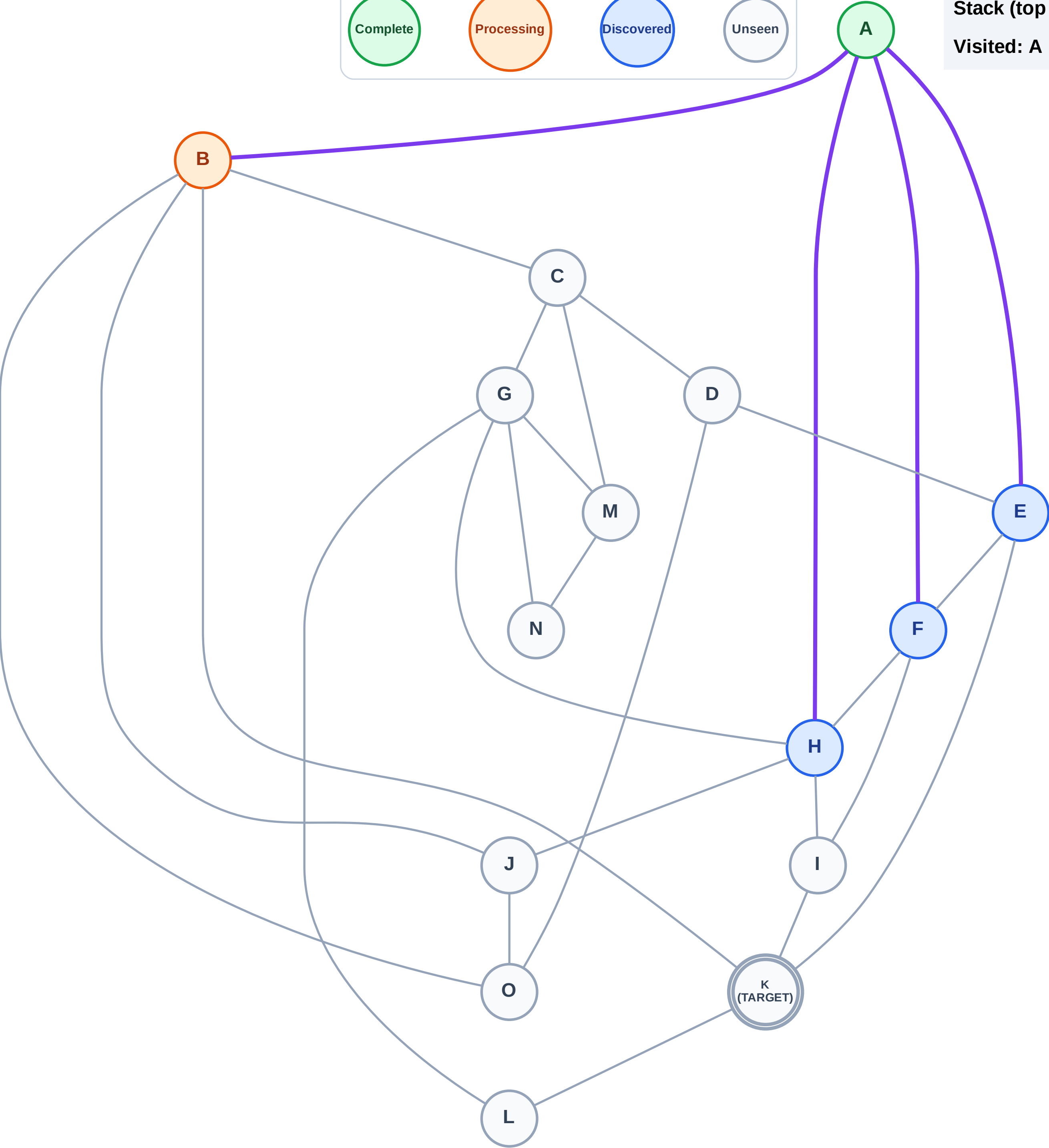
Processing

Discovered

Unseen

Stack (top at right): H | F | E

Visited: A



DFS Traversal

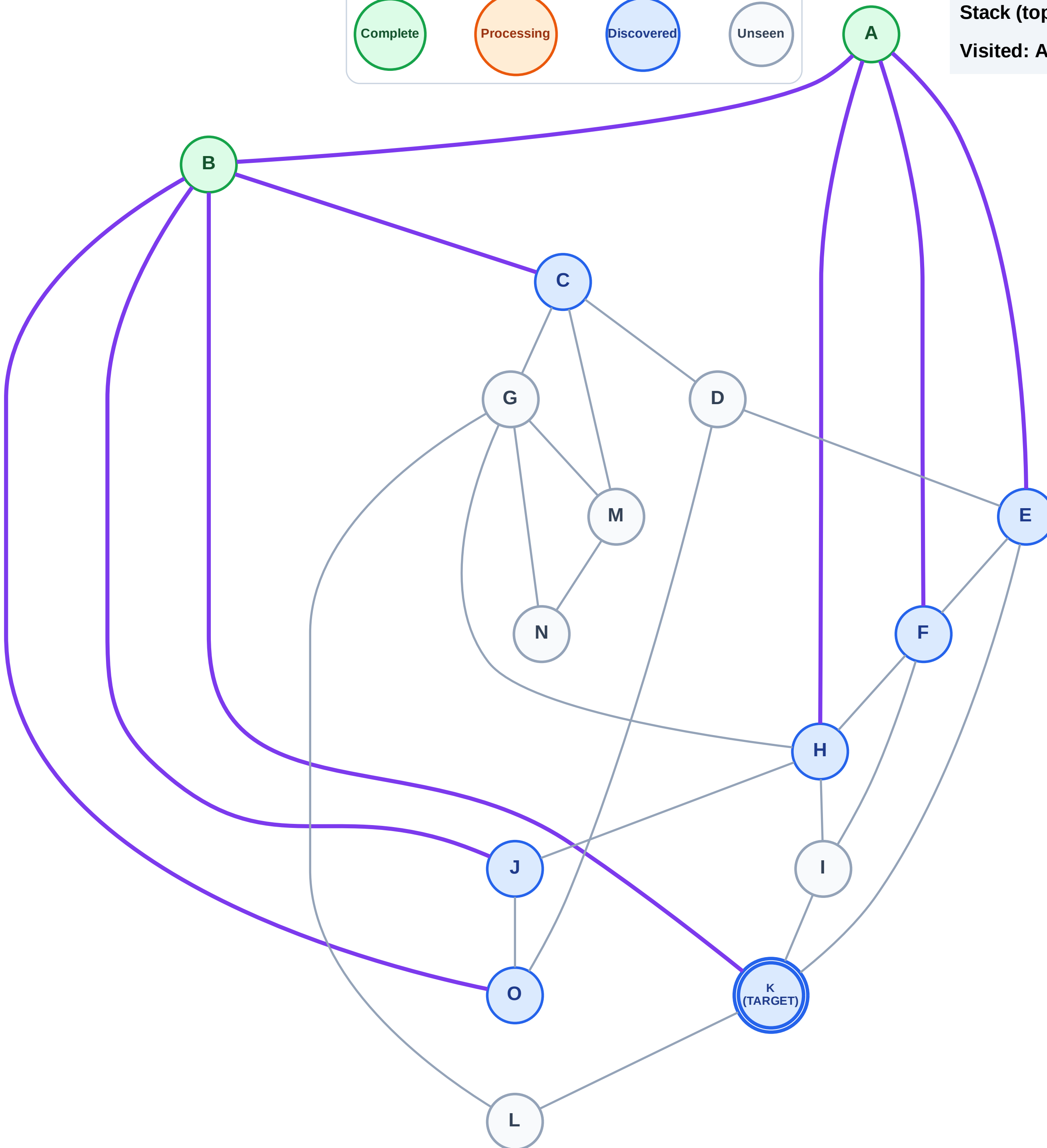
Step 5: Discover O, K, J, C from B

Legend



Stack (top at right): H | F | E | O | K | J | C

Visited: A, B



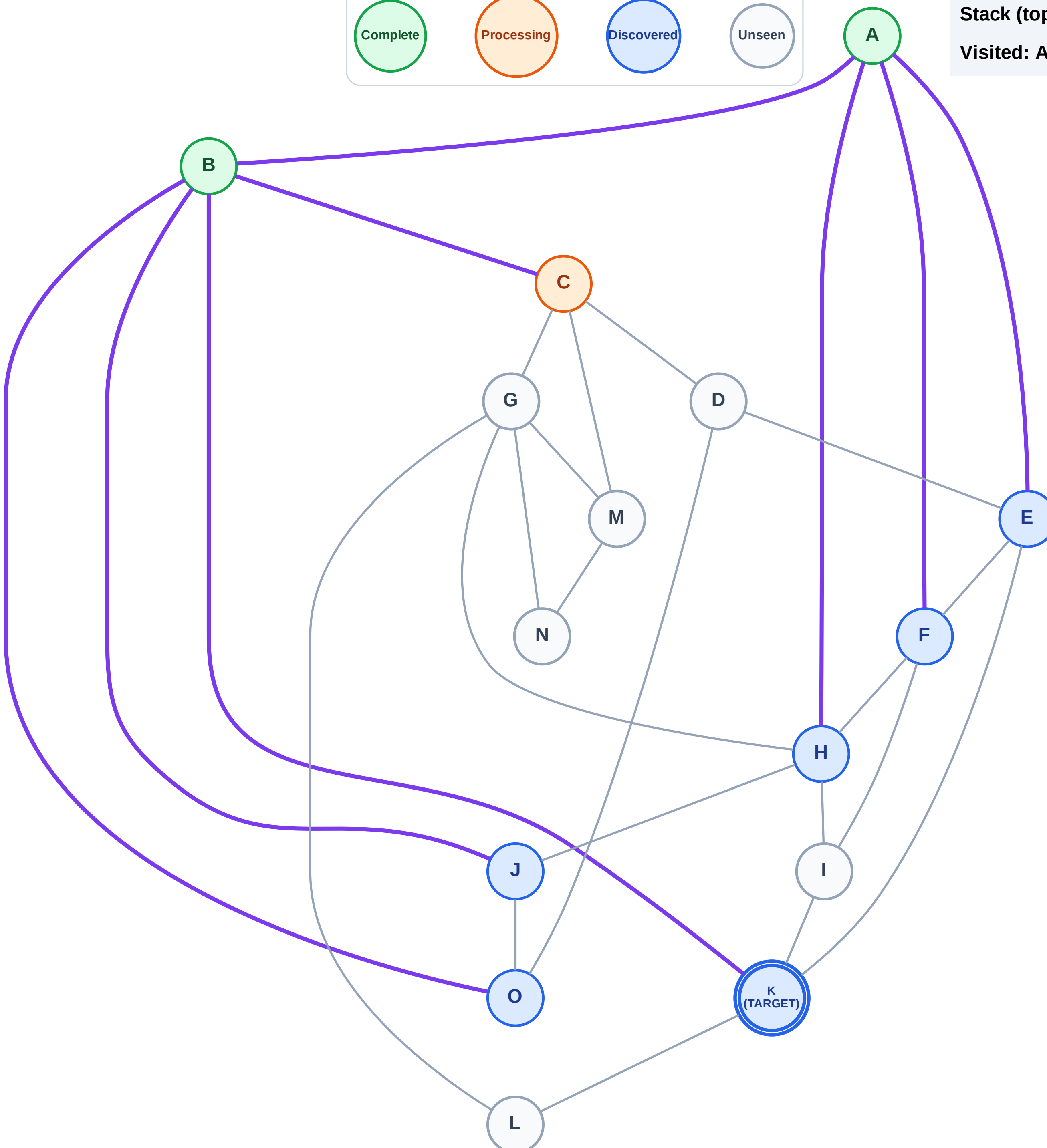
DFS Traversal

Step 6: Process node C

Legend



Stack (top at right): H | F | E | O | K | J
Visited: A, B



DFS Traversal

Step 7: Discover M, G, D from C

Legend



Complete



Processing

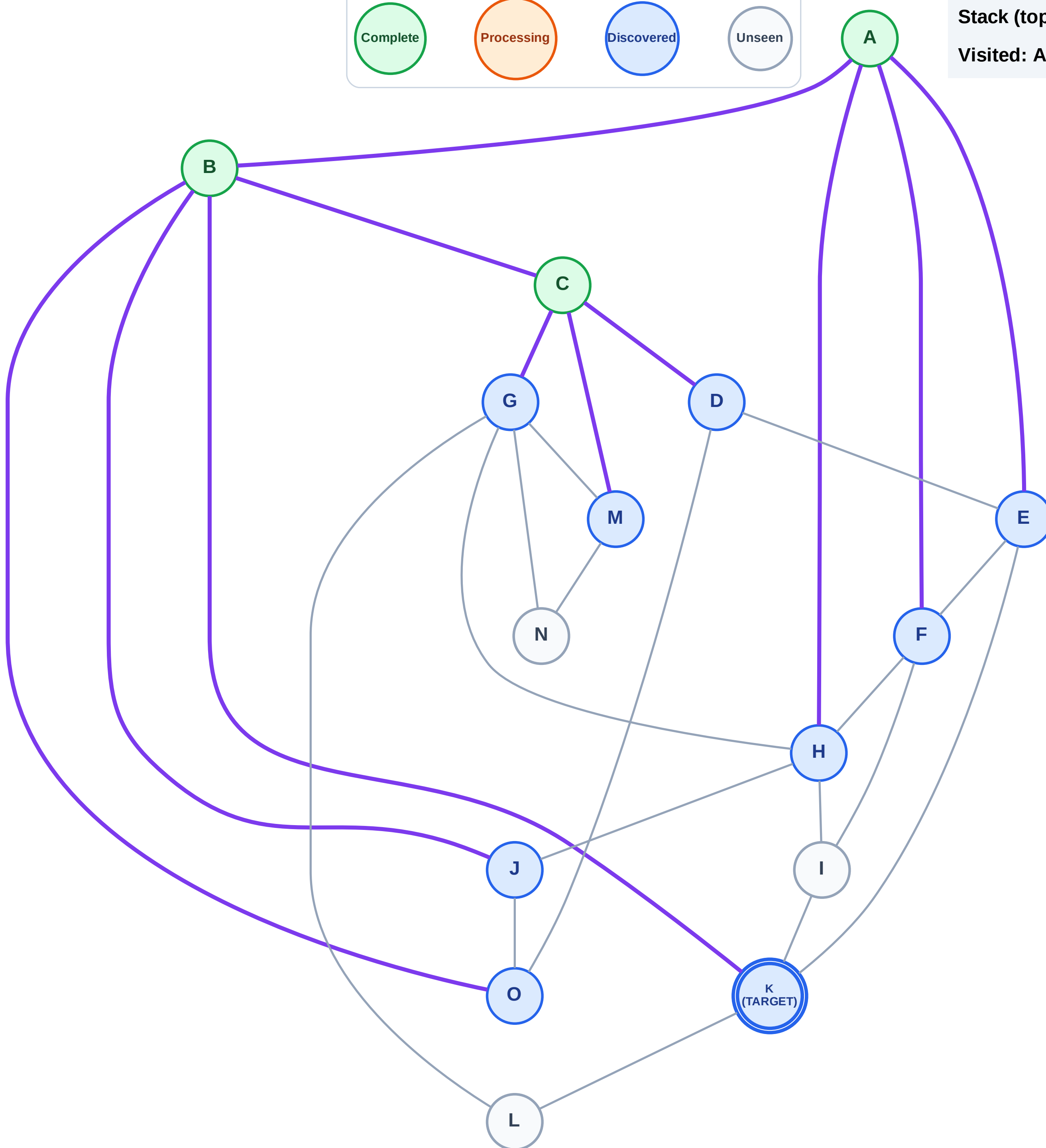


Discovered



Unseen

Stack (top at right): H | F | E | O | K | J | M | G | D
Visited: A, B, C



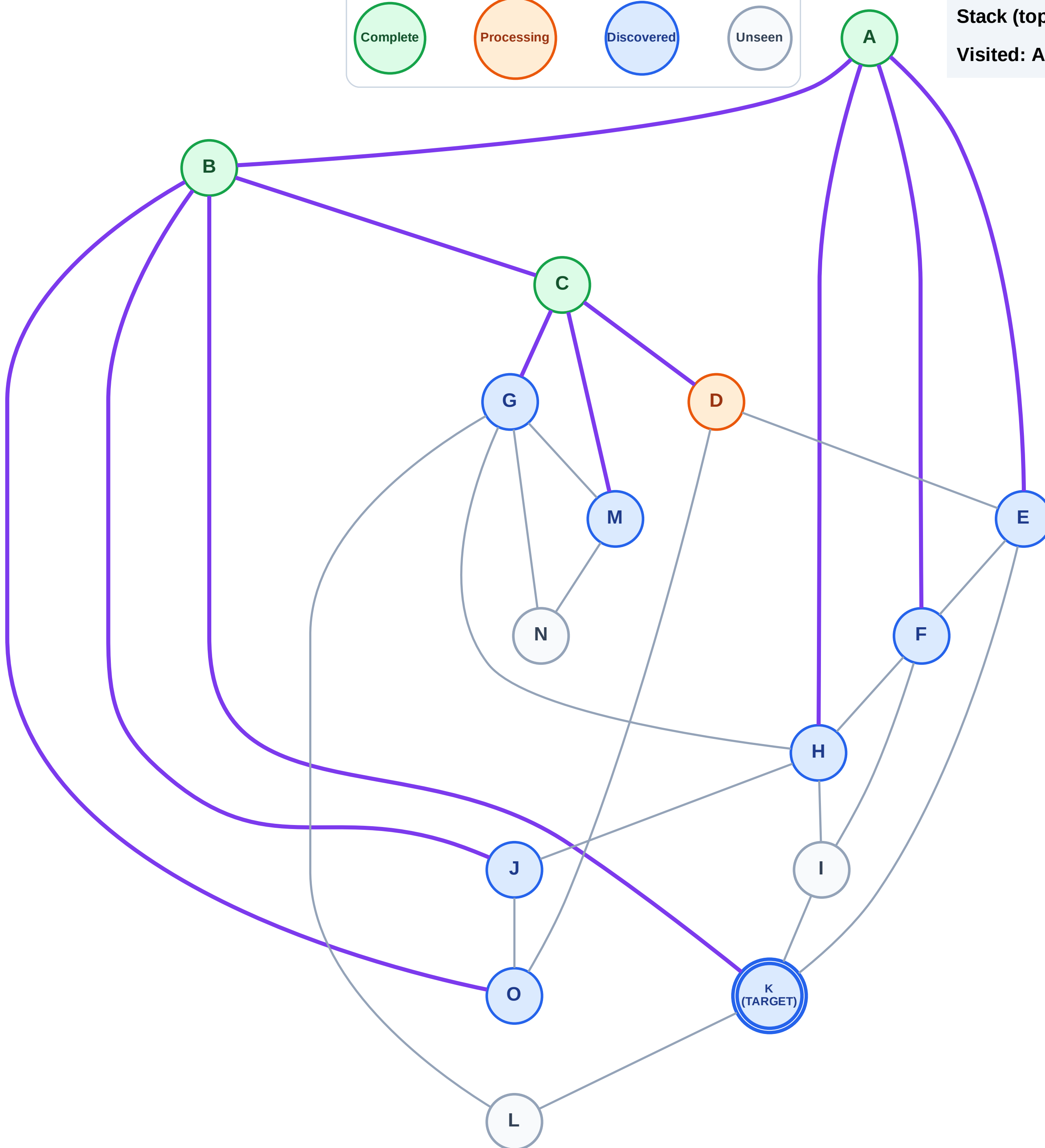
Step 8: Process node D

Page 10 of 10

Complete Processing Discovered Unseen

Unseen

Visited: A, B, C



DFS Traversal

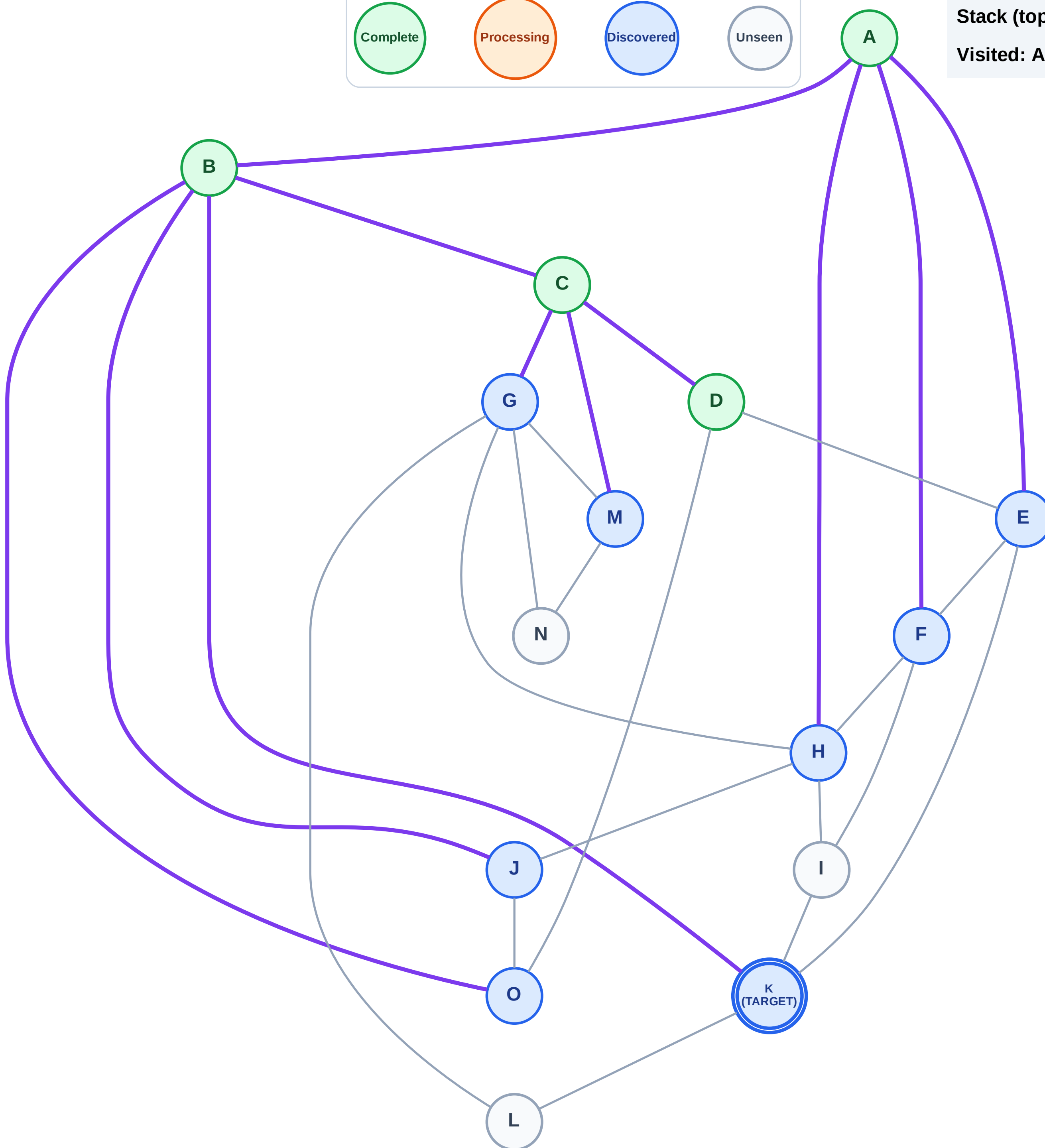
Step 9: No new neighbors from D

Legend



Stack (top at right): H | F | E | O | K | J | M | G

Visited: A, B, C, D

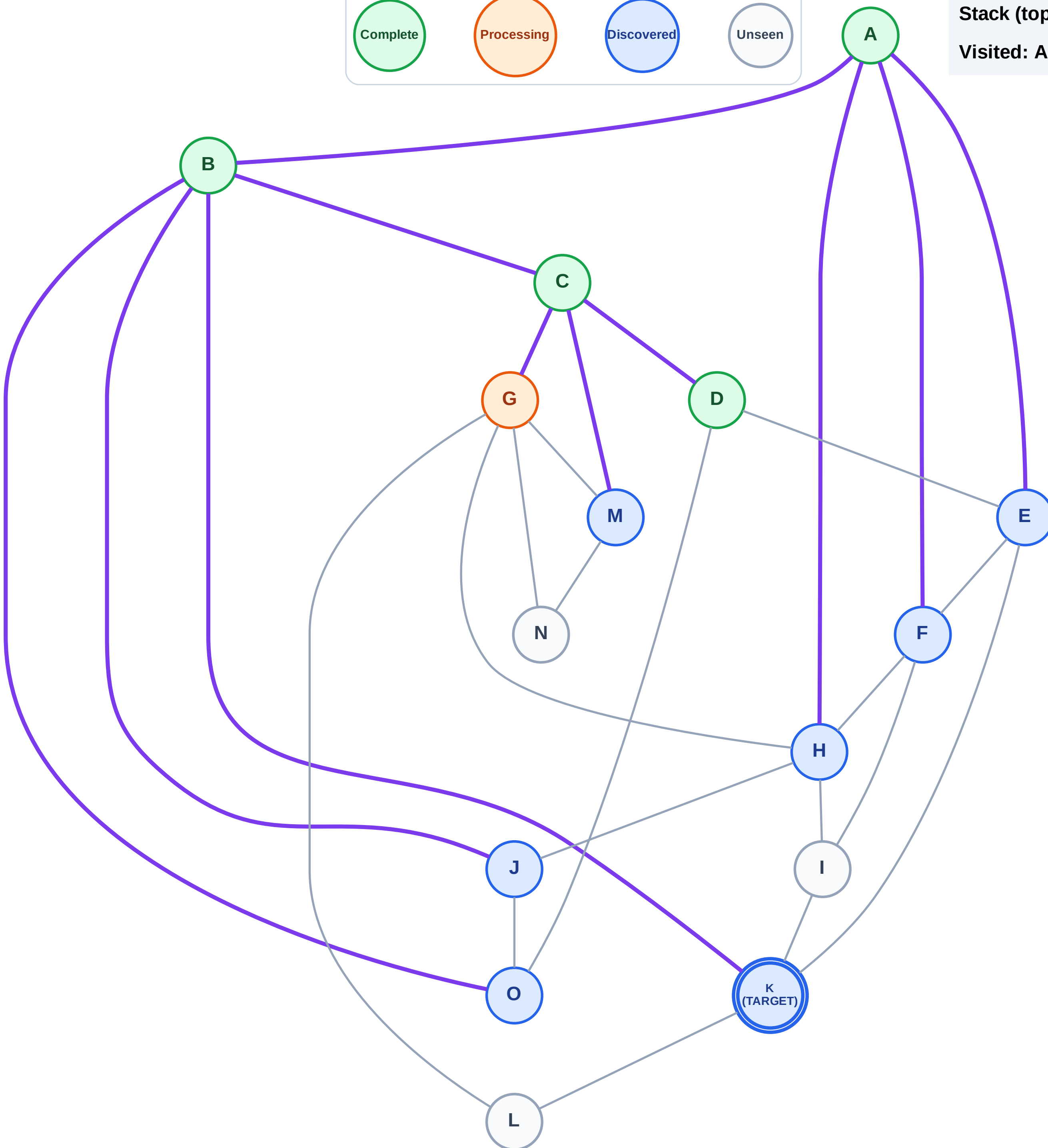


Step 10: Process node G

Legend

Unseen

Visited: A, B, C, D



DFS Traversal

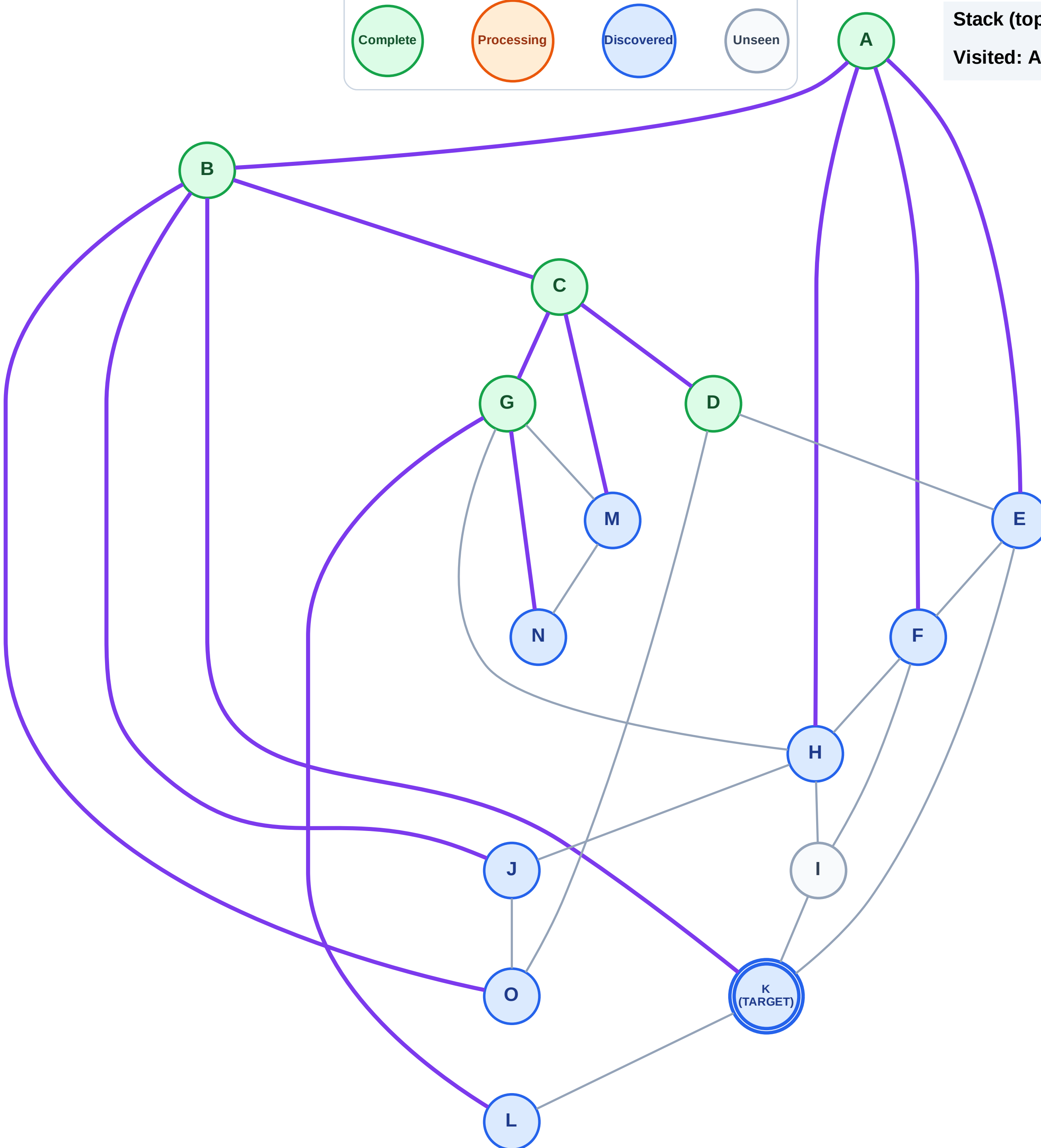
Step 11: Discover N, L from G

Legend



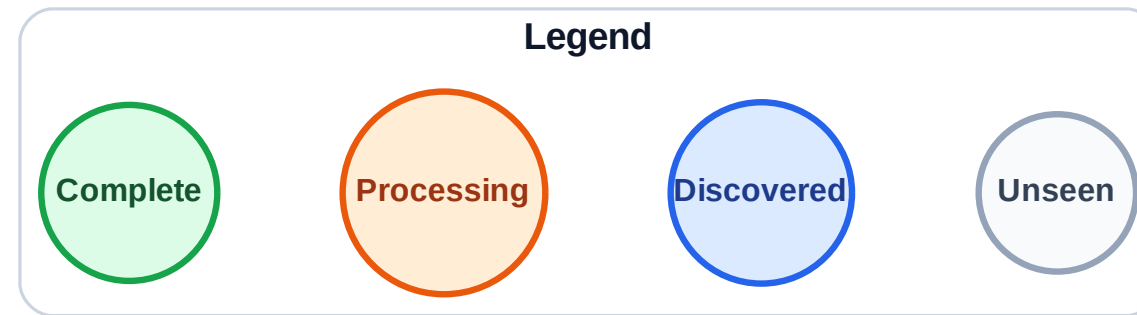
Stack (top at right): H | F | E | O | K | J | M | N | L

Visited: A, B, C, D, G



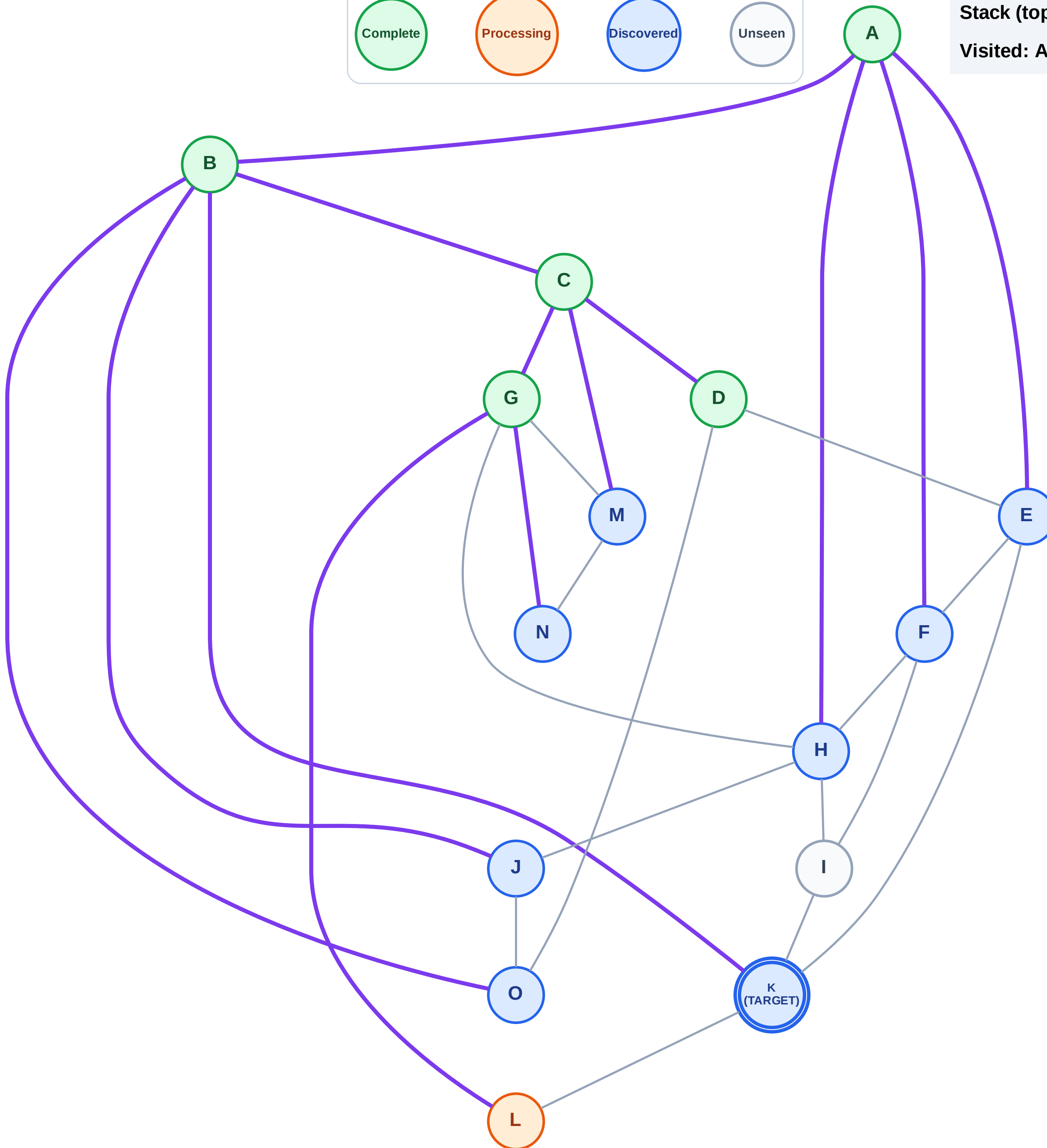
DFS Traversal

Step 12: Process node L



Stack (top at right): H | F | E | O | K | J | M | N

Visited: A, B, C, D, G



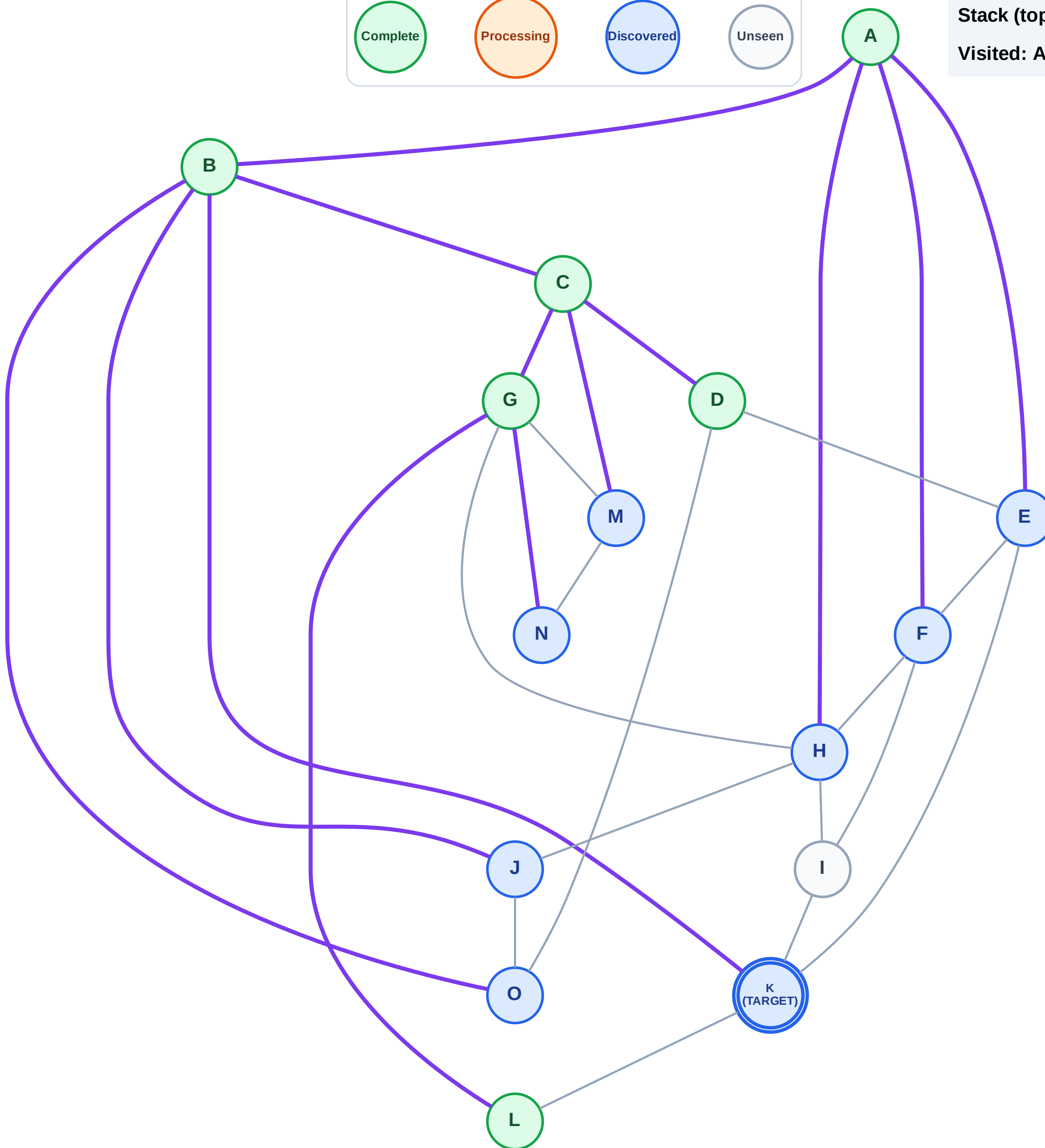
Step 13: No new neighbors from L

Legend



Complete Processing Discovered Unseen

Visited: A, B, C, D, G, L



DFS Traversal

Step 14: Process node N

Legend



Complete



Processing



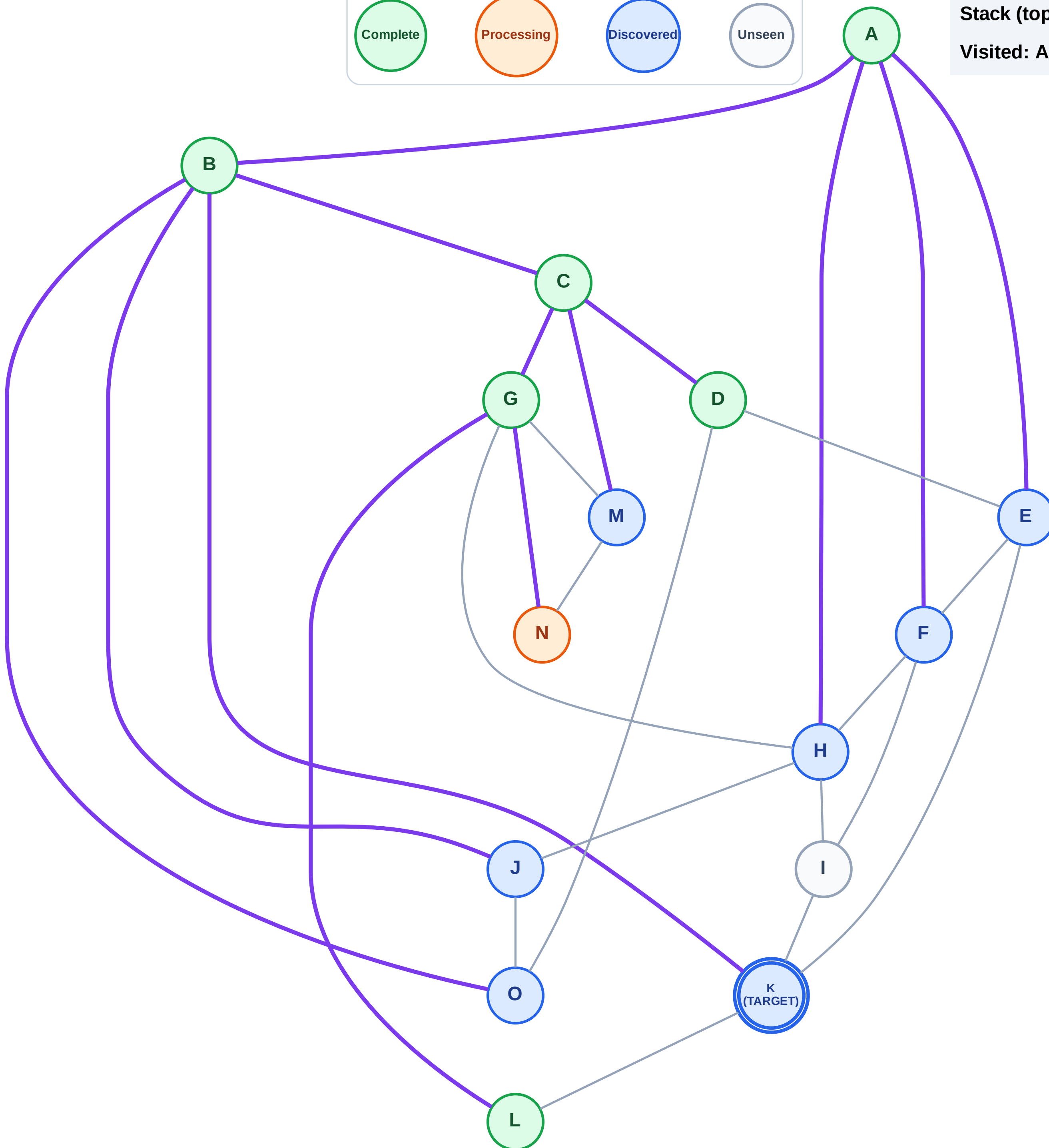
Discovered



Unseen

Stack (top at right): H | F | E | O | K | J | M

Visited: A, B, C, D, G, L



DFS Traversal

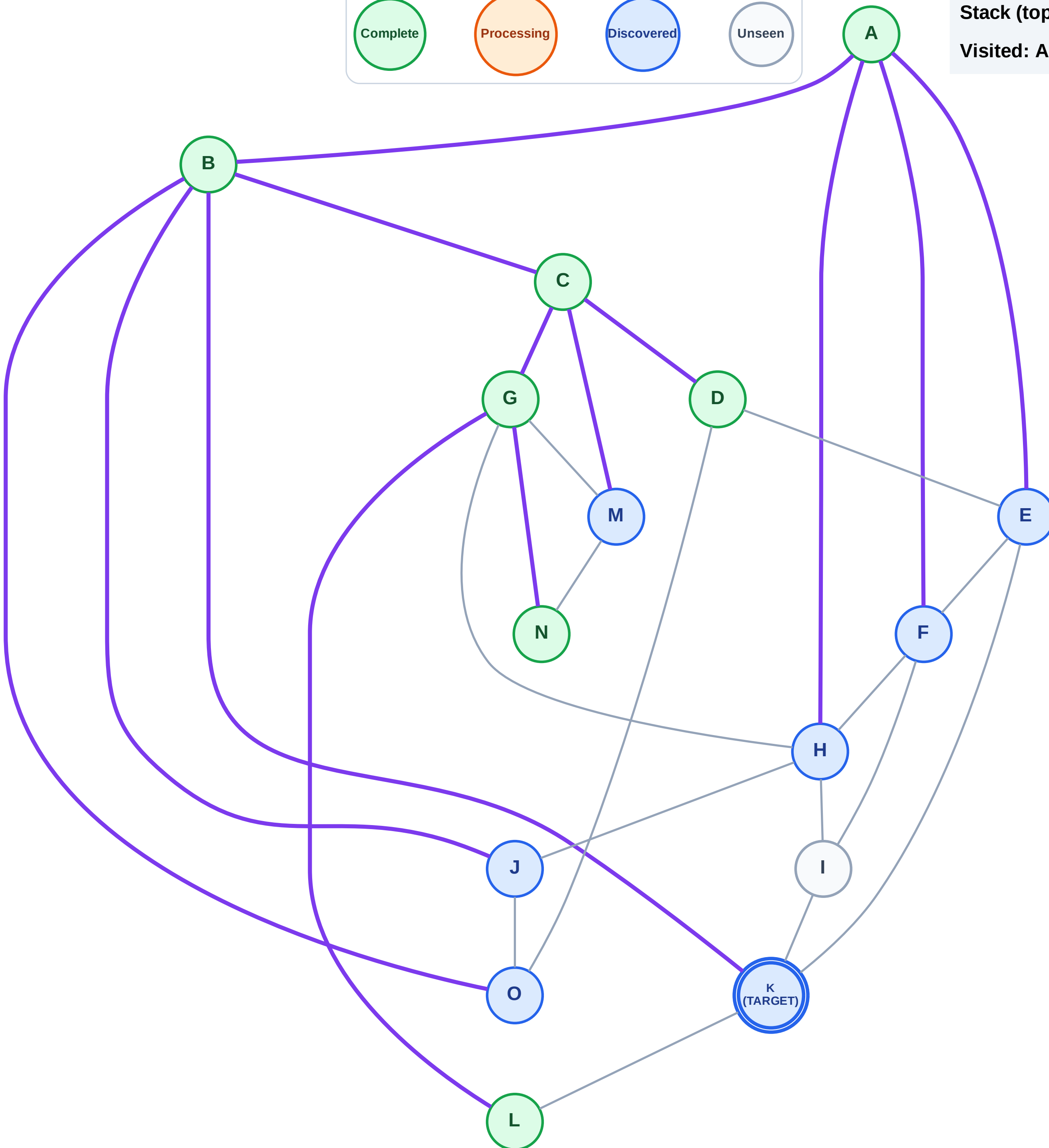
Step 15: No new neighbors from N

Legend



Stack (top at right): H | F | E | O | K | J | M

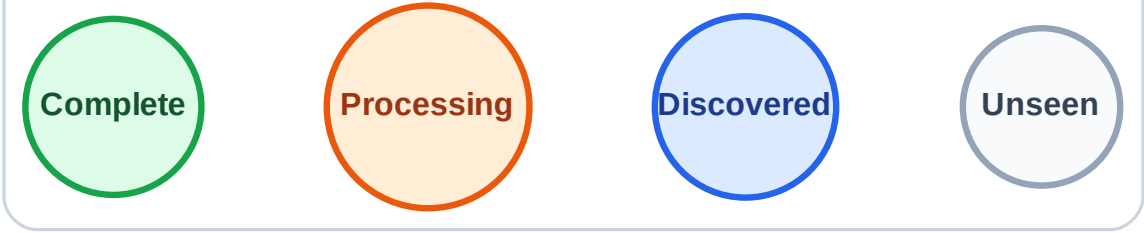
Visited: A, B, C, D, G, L, N



DFS Traversal

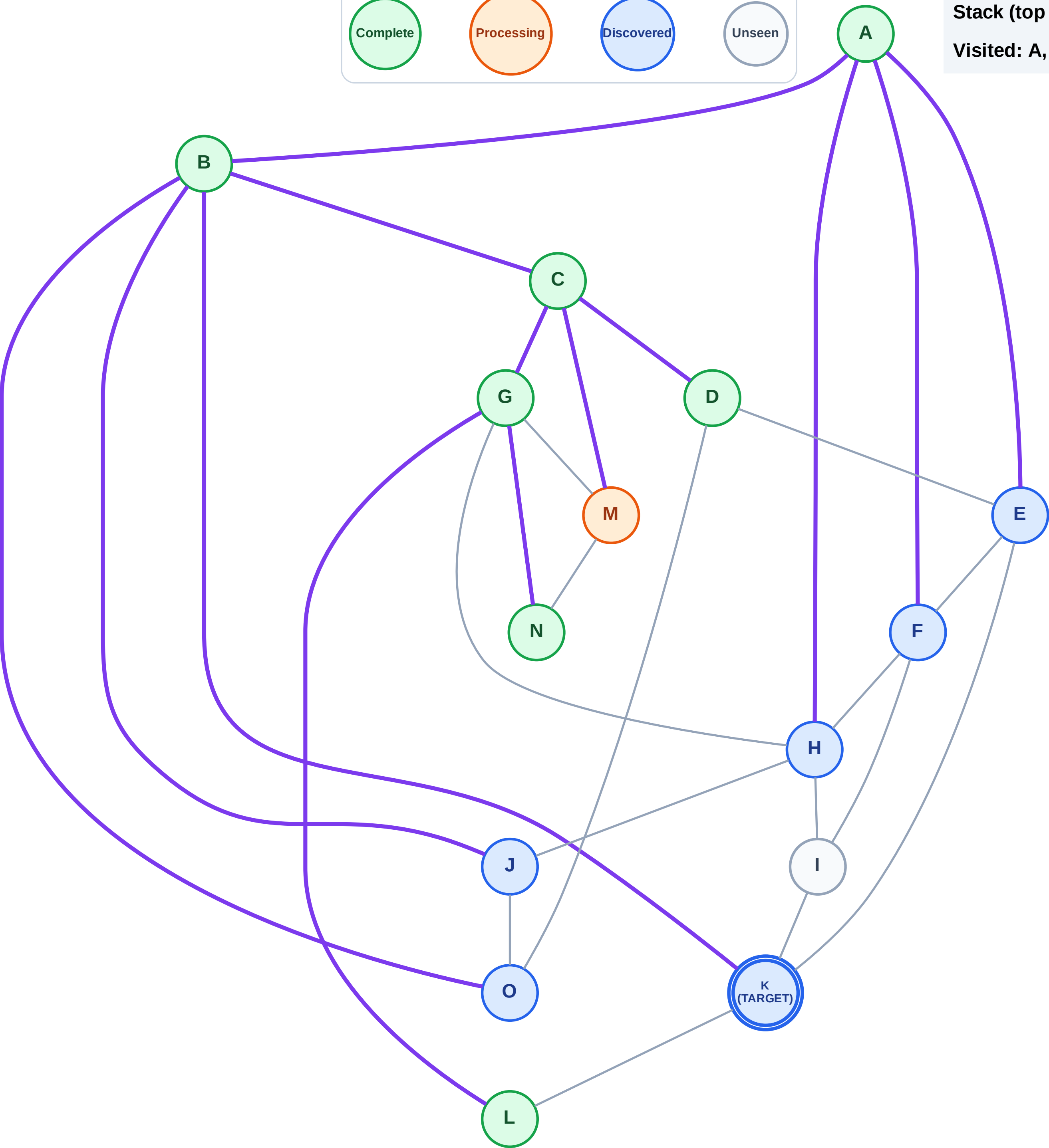
Step 16: Process node M

Legend



Stack (top at right): H | F | E | O | K | J

Visited: A, B, C, D, G, L, N



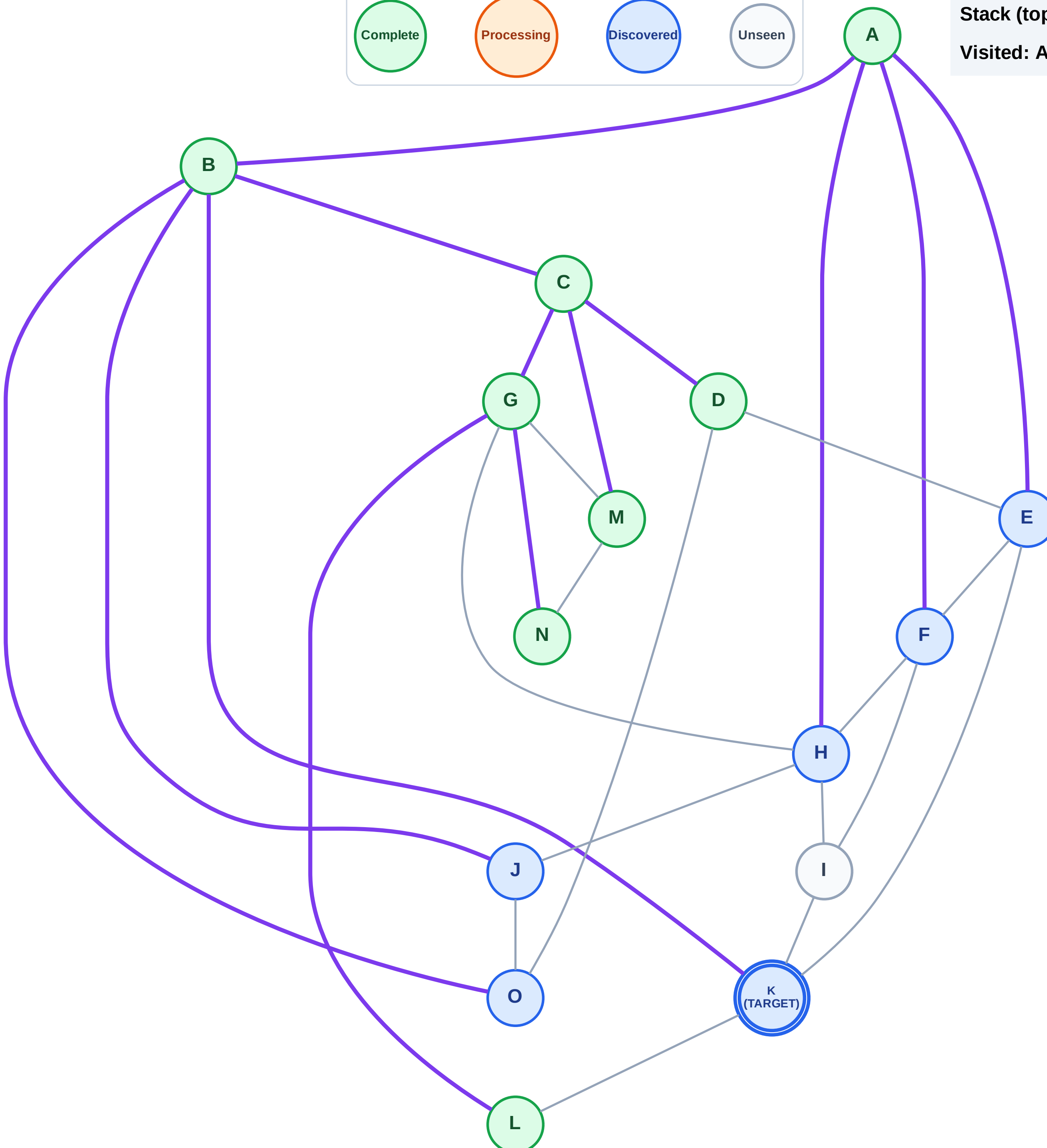
DFS Traversal

Step 17: No new neighbors from M

Legend



Stack (top at right): H | F | E | O | K | J
Visited: A, B, C, D, G, L, M, N



DFS Traversal

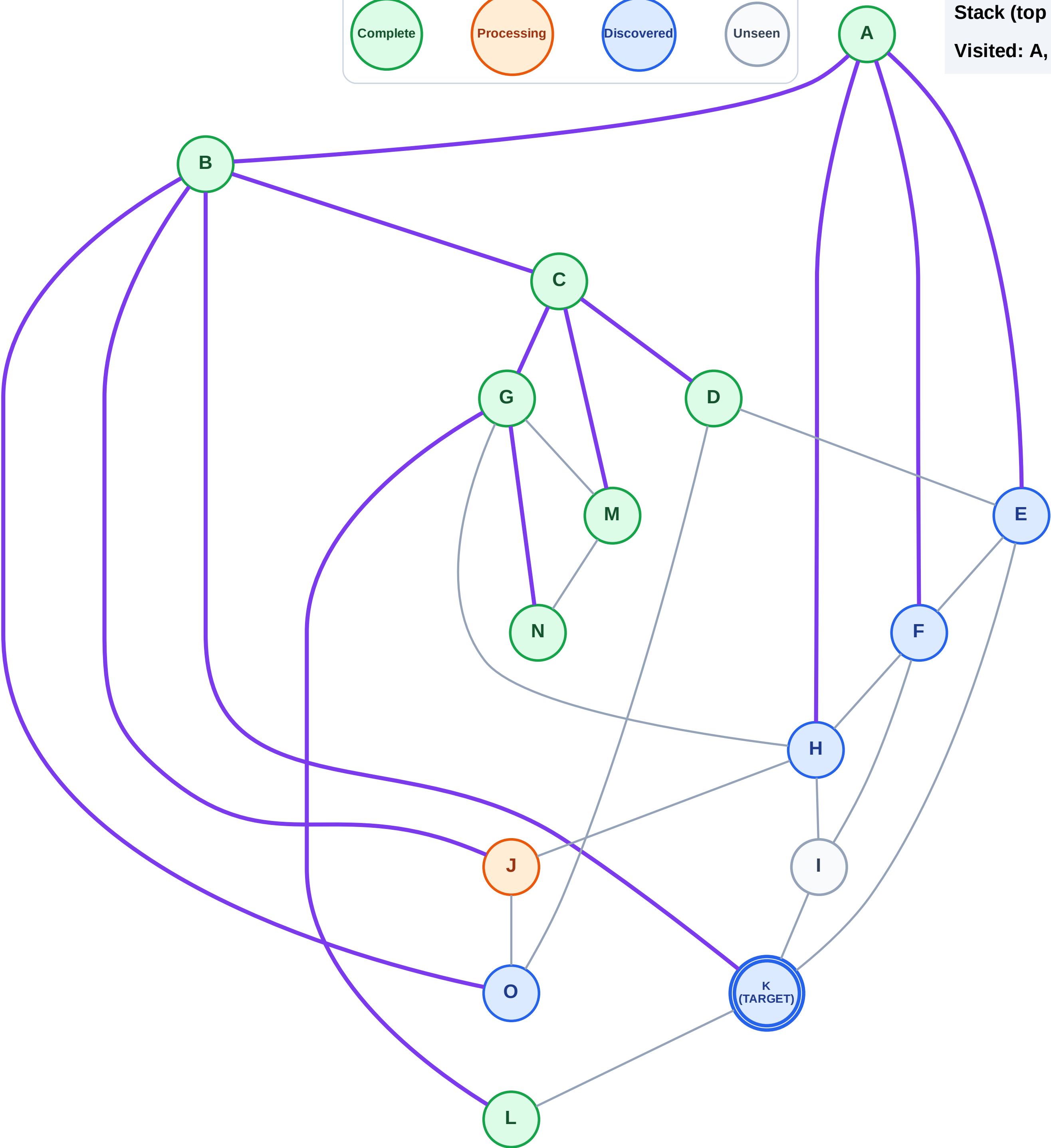
Step 18: Process node J

Legend



Stack (top at right): H | F | E | O | K

Visited: A, B, C, D, G, L, M, N



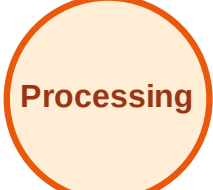
DFS Traversal

Step 19: No new neighbors from J

Legend



Complete



Processing



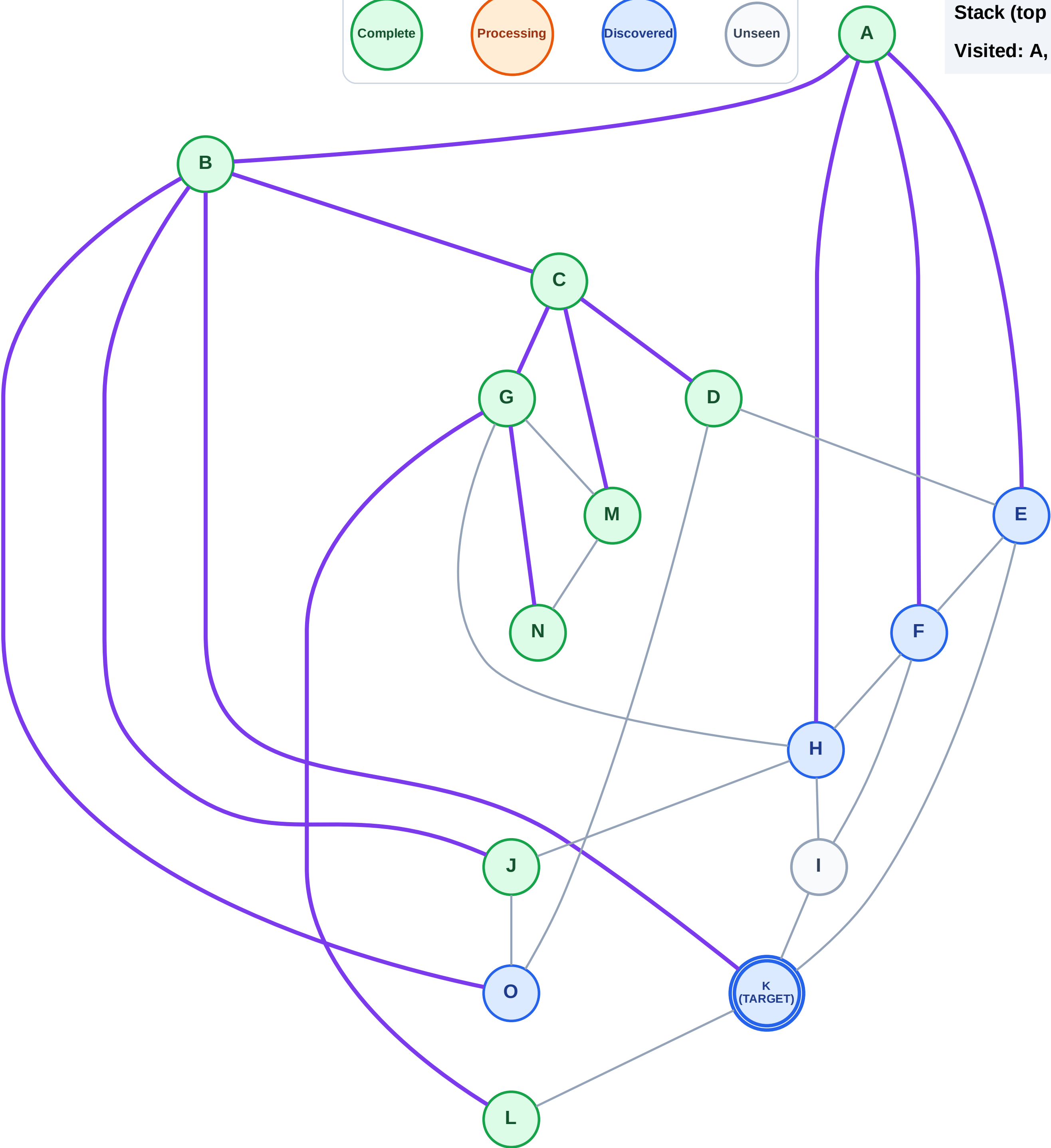
Discovered



Unseen

Stack (top at right): H | F | E | O | K

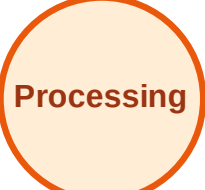
Visited: A, B, C, D, G, J, L, M, N



DFS Traversal

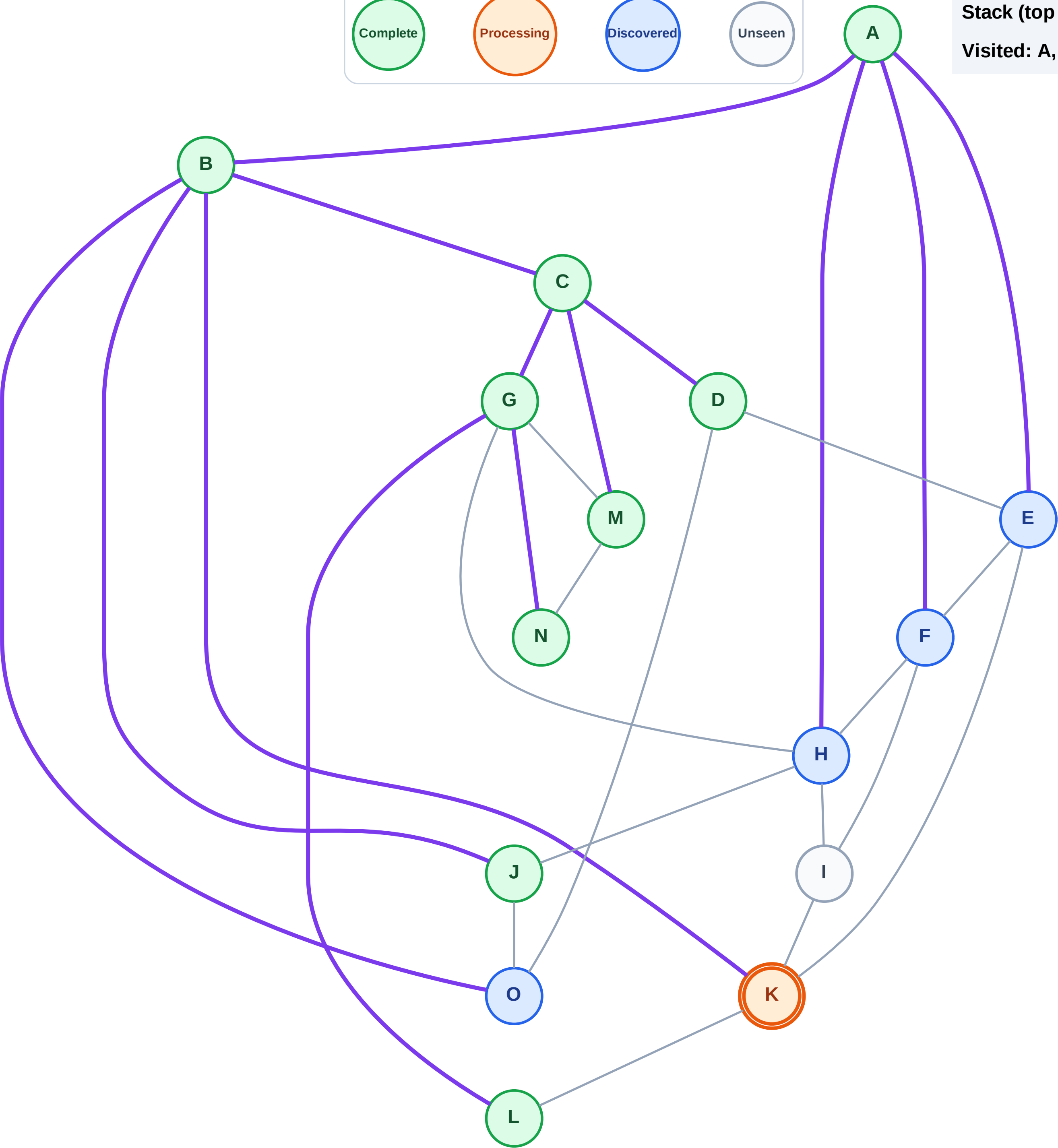
Step 20: Process node K

Legend



Stack (top at right): H | F | E | O

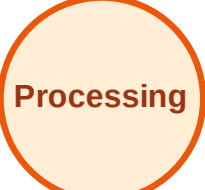
Visited: A, B, C, D, G, J, L, M, N



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Stack (top at right): H | F | E | O

Visited: A, B, C, D, G, J, K, L, M, N

